8

Current Trends in Artificial Intelligence

8.1 AI and Ethical Concerns

We know that ethics define the discipline that deals with moral obligations and duties of humans.

Correspondingly, ethics of AI deals with ethics specific to robots and other artificially intelligent beings. These ethics can be divided into two groups.

1. Roboethics: Roboethics consider moral behaviour of humans as they design, construct, use and treat artificially intelligent beings.
2. Machine ethics: These ethics manage the moral behaviour of artificial moral agents (AMAs) that play an important role in delivering artificial general intelligence (AGIs) within existing legal and social frameworks.

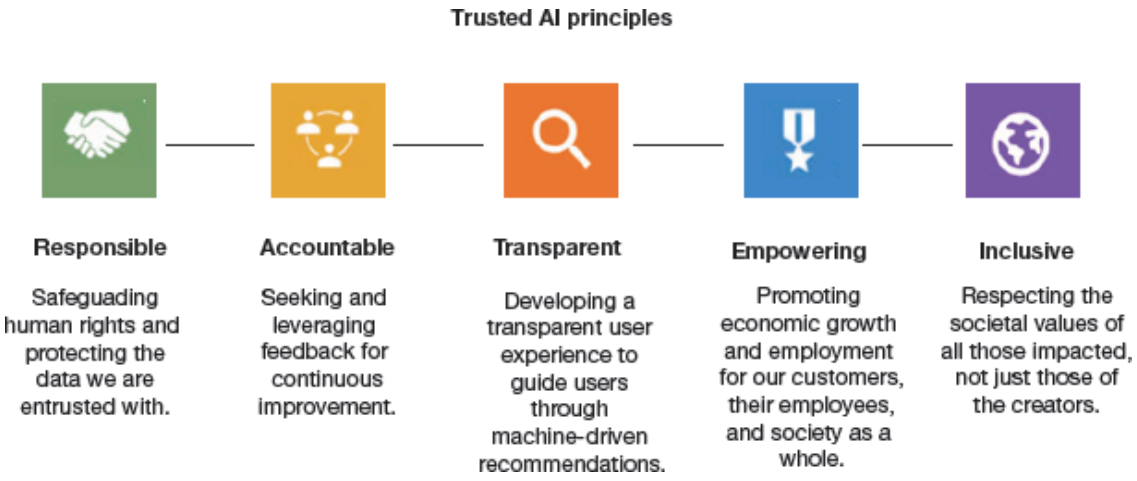


FIGURE 8.1 Trusted AI principles

The bigger concerns are as follows:

1. If AI generates human-like output, can it also make human-like decisions?
2. If AI makes human-like decisions, then are these decisions human-like?
3. If AI takes decisions of sanctioning or rejecting a bank loan, is that decision justified?
4. If AI decides whether a student should be enrolled in a college or not, is AI taking a fair decision?
5. If AI makes a human-like decision, is it human-like trustworthy?

AI is basically data + mathematical model + training based on data + predictions. What if the data provided for the training is unintentionally wrong/biased?

Issues like these are endless. Hence, getting a good understanding of them is critical to the success of this technology. Refer **Fig. 8.1** to learn about some guiding principles of a trusted AI Tool. As a citizen of the AI society, we must be aware about how AI works and the framework of AI ethics.

8.1.1 Ethical Use of Artificial Intelligence

AI systems no doubt offer a wide range of functionality for businesses, but their use also raises ethical questions.

An ideal AI system must not be biased. This is especially true when AI uses inherently unexplainable deep learning and generative adversarial network (GAN) algorithms.

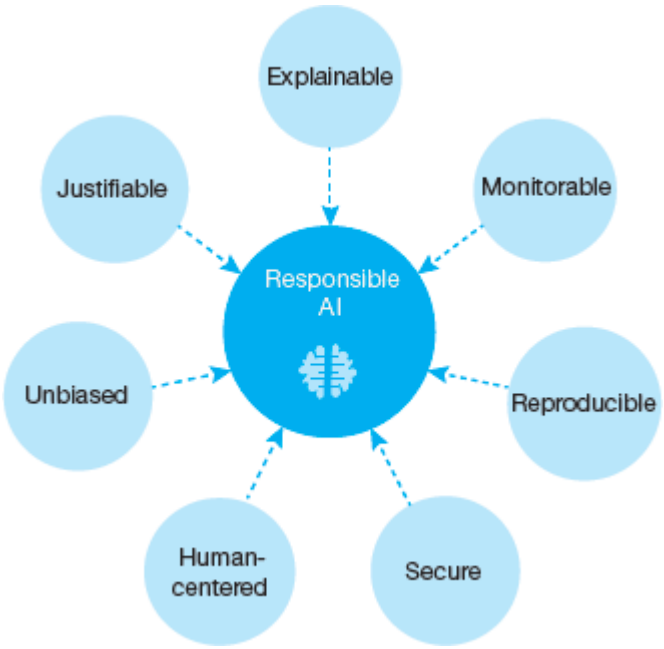


FIGURE 8.2 Features of Ethical AI System

The ability to explain is a potential stumbling block to using AI systems in industries that operate under strict regulatory compliance requirements. For example, financial institutions must explain their credit-issuing decisions. The main problem with such systems is that it will be difficult to explain how the decision was taken to refuse a credit as such decisions are made by teasing out subtle correlations between thousands of variables. In such a scenario, the AI program that is not able to explain decisions is known as black box AI.

A responsible AI system is explainable, monitorable, reproducible, secure, human-centered, unbiased and justifiable (refer **Fig. 8.2**).

Formulating laws to regulate AI is not at all easy as AI uses multiple technologies. Any strict law can hamper its progress and development. The AI technology is changing so rapidly that new technology breakthroughs sometimes make existing laws obsolete. For example, existing laws regulating the privacy of conversations and recorded conversations do not cover the challenge posed by voice assistants like Amazon's Alexa and Apple's Siri. These digital assistants collect data not to distribute them but to improve its machine learning algorithms. This is a golden chance for criminals having malicious intent.

8.1.2 Is AI Dangerous? Will Robots Take Over the World?

Rapid growth and increase in the capabilities of AI systems have made us wondering about the risks involved in using AI. Some of us think that AI will quickly takeover tasks performed by humans. But most researchers believe that super-intelligent AI is unlikely to exhibit human emotions, so there is no point thinking that AI will become malevolent. AI can become a risk only in two conditions.

AI system is specifically programmed to do something devastating. For example, autonomous weapons are AI systems that are programmed to kill. If acquired by scrupulous people having ill intentions, such weapons could inadvertently lead to an AI war, and mass casualties, potentially even the end of humanity.

Autonomous weapons are designed to be extremely difficult to 'turn off', and once they become operational, humans could themselves rapidly lose control over them. This risk is also present with narrow AI when given autonomy.

Just imagine what would happen if an autonomous drone with facial recognition as well as a 3D-printed rifle, pistol or other gun becomes available. Or if a self-driving car, connected to the Internet is hacked to get into some serious accident.

Even in hospitals, more and more equipment are now connected to the Internet. If any or all of these are hacked, then can you imagine what the hacker can do with the patient's body?

AI could be programmed to do something beneficial, but the method used to achieve its goal can be highly destructive. This is especially true when the AI programmer asks the machine to complete a task, but does not clearly outline the goals. For example, you can instruct a self-driven car to program to take you to a particular destination as soon as possible. However, the instruction 'as soon as possible' fails to address safety, road rules, etc. The smart car may successfully complete its task, but after creating a great havoc. So, there must be some provision to continuously

monitor and control the machine, its goals and process being performed to achieve that goal.

Of 9 million low-skilled services and BPO roles, 30 per cent or around 3 million will be lost by 2022, principally driven by the impact of robot process automation or RPA.

As of now, there is no liability for actions on machines. There is no clarity on what legal aspects bind machines when they become increasingly smart. The following questions may arise:

1. Do we judge AI systems the same way as we judge a human?
2. Who is responsible if the AI system becomes self-learning and autonomous to a greater extent?
3. Is there any error margin for AI machines, even if it has fatal consequences?

Apart from these two serious concerns, AI also poses some additional threats/risks that calls for attention. These threats are discussed below.



Credit: PaO_STUDIO / Shutterstock

FIGURE 8.3 Effect of AI on jobs

1. **The immediate risk posed by job automation:** As AI robots become smarter and more dexterous, they will quickly replace humans. For example, you must have heard of restaurants having robots as wait-

ers. Just imagine, where the less educated people will go if machines start taking their jobs (as seen in **Fig. 8.3**)?

It is often rightly feared that the deployment of AI systems for job automation will eventually replace certain types of jobs, especially those that are predictable and repetitive. According to a 2019 Brookings Institution study, 36 million people work in jobs that they may soon lose owing to automation as at least 70 percent of their tasks (varying from retail sales, market analysis to hospitality and warehouse labour) will be done using AI. In fact, a newer Brookings report even states that white collar jobs may be worst affected. As per McKinsey & Company report (2018), the African American workforce will be hardest hit by automation.

2. **Biased algorithms:** We know that computers work on GIGO concept which says Garbage-In-Garbage-Out. This is a very serious limitation in AI applications. If we feed our algorithms data sets that contain biased data, then the output generated from such systems will only produce biased results. This biased result, if applied to solve a real-world problem, may lead to an even bigger problem.

   A dynamic advertising billboard in Utrecht was switched off because the spy software installed on these billboards aroused public outrage.
3. **Too little privacy:** When using IoT or AI systems, a lot of data is generated. It is said that approximately 2.5 quintillion bytes (or 2.5 million terabytes) of data is added each day. It is interesting to note that 90% of the digital data has been created in the last two years. A lot of data is required for the proper functioning of the smart systems. As a result, much data about us is collected thereby eroding our privacy. Once the systems collect our data, we have no means to know what data about us is used by whom and for what purpose.

   These days, cameras can easily be fitted with facial recognition software to capture details about us (including our gender, age, ethnicity, gesture and state of mind). Do you know that in China, some police officers wear glasses with facial recognition technology having a database with facial pictures of thousands of ‘suspects’ (judged on the basis of certain behaviour)?

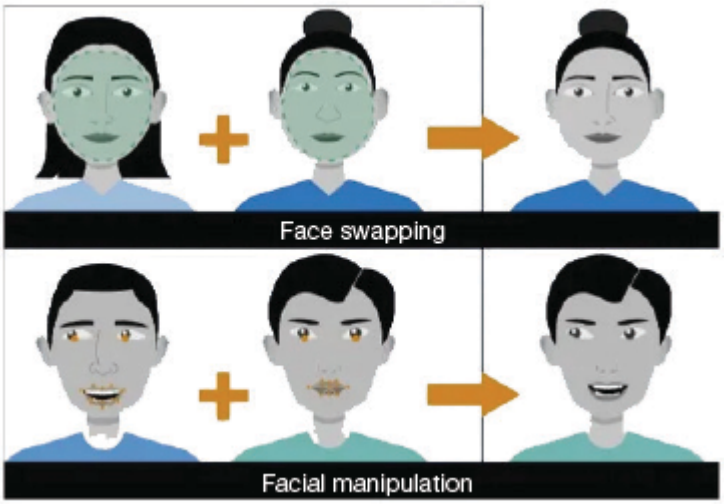


FIGURE 8.4 AI Tools can be used for Face Swapping and Face Manipulation

4. **Everything becomes unreliable:** These days, fake news and filter bubbles are not uncommon. Smart systems can create faces, compose texts, produce tweets, manipulate images, clone voices and engage in smart advertising.

An AI system can be used to depict day into night or create highly realistic faces of people who have never existed.

For example, open-source software Deepfake can easily stick pictures of faces on moving video footage. Due to such fake videos, celebrities are the worst affected. People with malicious intentions easily create pornographic videos starring them. Even normal citizens are blackmailed with morphed images and/or videos. This is often known as **Faceswap video blackmailing** (refer **Fig. 8.4**).

Even in politics, fake or manipulated videos are produced at a high speed to influence the opinions of the masses.

Did you know that the company Cambridge Analytica, was found to manage access to data from 87 million Facebook profiles of Americans to campaign for President Trump so that he could come to power?

8.1.3 Ethics in AI

The current world is quickly getting transformed by artificial intelligence and those who use this technology are living a richer and easy-going life. However, this is possible only when we talk about ethical AI, not just AI.

The term, ethical AI, is composed of two words—ethical and AI. We are now already familiar with the term AI. Ethical means one that is according to ethics. And ethics relates to moral principles that govern the behaviour and actions of a group or an individual. So, ethical AI is the theory behind developing computer systems that can perform tasks that require human intelligence. It focusses on how right, how fair and how just is the AI's output, outcome and impact.

Example: We know that AI can produce biased results if it is trained with biased data or if it does not do anything to avoid bias. So, any AI system that works on biased data is actually not ethical (or unethical). For example, Microsoft released a chatbot called Tay on Twitter in 2016, to learn by engaging people in dialogue through tweets or direct messages. However, trolls made Tay learn all negative words that spread hatred for women, a particular race and all sorts of biased data within hours. Tay was also taught to give repulsive and toxic responses. This aroused the anger of Twitter users and finally Microsoft had to silence Tay forever in even less than 24 hours.

An ethical AI system designed to solve a particular problem, uses unbiased data and is trained using the right learning model for the problem and is monitored to evaluate if the results produced by it is right and fair.

8.1.4 AI and Bias

Think about the following statements.

1. Most images that show up when you do an image search for 'doctor' are white men.
2. Images and videos of a 'doctor' show usually shows a male character and that of a 'nurse' is a woman.

3. All virtual assistants like Alexa, Siri, Google assistant, etc. work with female voice.
4. Computer vision systems frequently give an error while recognizing people of different colour.

Can you think of more such cases where we witness biasness towards a particular gender, colour, caste, religion, country, etc? Technically, AI bias occurs when an algorithm produces results that are systematically prejudiced towards a particular group of people and thus produces skewed output. AI applications may have built-in biases because ultimately, they are created by humans who have conscious or unconscious preferences that may go undiscovered until the algorithms are used at a massive scale. Let us understand three basic sources of bias in AI systems.

Data

Any AI system is just as good as the data we put into them. Naturally, adding biased or skewed data can never give fair results. Machines do not have the intelligence to differentiate between right or wrong training data.

For example, many-a-time, you must have noticed that Google or any other voice assistant replies in American accent and at times gives wrong results when we give a command in Indian accent. This happens because the AI system is trained on recorded speech from white, upper-middle class Americans, making it difficult for the technology to understand commands from people not belonging to this category.

Algorithm

Algorithm can further amplify the biases injected by skewed data. For example, if you Google Image Search ‘Teacher’, then the image classifier algorithm trained on the images available in public domain shows more women teaching as opposed to male teachers. AI algorithms must be designed to maximize accuracy. But in this case, the AI algorithm may decide that all people teaching are women, despite the fact the training data has some images of men in the classroom. So, this is a case of gender bias in the AI system that becomes a part of AI algorithm due biasness in data.

People

People who are developing the AI system, that is, engineers, scientists, developers, etc. are the next big source of bias. Humans design AI systems to give the most accurate results with the available data. So, data feeders that may be humans may again hit the success unknowingly. ***Therefore, it is rightly said that ethics and bias are not the problem of the machine but that of the humans behind the machine.***

8.1.5 Towards Ethical and Trustworthy AI

Today, several companies worldwide are using algorithm-based 'emotional AI' hiring platforms to augment and lower the financial burden of their recruitment processes. While these systems may hire in a fair and unbiased manner, there are cases reporting women applicants being disproportionately rejected based on years of biased data in a male-dominated sector.

Let us now see few ways in which world is trying to deal with this situation.

Regulating a More Ethical AI

In April 2021, the European Commission launched its first ever legal framework on AI to guarantee the safety and fundamental rights of people and businesses, while strengthening AI uptake, investment and innovation across the EU.

The new risk-based approach will not only set strict requirements for AI systems but also immediately ban AI systems which are considered to 'be a threat to the safety, livelihood and rights of people' including 'systems that manipulate human behaviour, circumvent users' free will and allow social scoring by governments'.

Later, in June 2021, Australia launched a similar AI ethics framework to guide businesses, governments and other organizations to responsibly design, develop and use AI.

Company and Organizational Engagement

Besides necessary regulation and standards, these days, organizations are building trust in AI technologies through cultural and educational programmes, risk assessments and third-party audit programmes. This would help them to move towards a prevention, detection and response framework (like anti-corruption, prevention of tax evasion frameworks).

Rights and Activist Groups

AI systems are yet to go a long way. In this journey, the relationship between people and technology will be redefined continuously. To ensure that our societies move forward on the path of ethics, it is critical that AI systems are human-centric. Civil rights and activist groups need to challenge the discourse and amplify the voices of those who are most adversely affected by new technologies.

We must all ask the question—Is the technology necessary or is there any alternative? Only if the benefits outweigh the harm, we must check form whom the AI systems work.

Ensuring Data Privacy

Companies are now taking measures that are important to keep data safe and protect it from potential misuse. Privacy protection is now becoming an integral part of any ethical AI system. Data ethics and privacy in such systems is ensured by the following:

1. Encrypting the data.
2. Ensuring that the AI algorithm cannot learn from outside its dataset.
3. Using secure computation so that even the people who developed or work on those AI systems are not able to see or access the data.
4. Restrict reverse engineering the AI model so that no user can access data used for model training.

Diversify Your Team

Since bias is an important issue and results from an AI tool will be used by people from different racial, gender and economic identities, we need

to make sure that there should be no bias. For this, build diverse teams to reduce the potential risk of bias falling through the cracks. A diverse team has data scientists, business leaders, professionals with different educational backgrounds and experiences, such as lawyers, accountants, sociologists and ethicists. This would help the company to get their views on any sort of bias so that it could be mitigated in a timely manner. The team should continually analyse data and algorithms ensure fairness. There are many tools like **Bias Analyzer** which automate this process and also show the costs and benefits associated with a variety of possible mitigation actions.

8.1.6 Why is Ethical AI Important?

Since AI is used in areas like medicine, law enforcement, recruiting, data privacy, military defense or self-driving vehicles, it makes it mandatory that the AI system must produce accurate, transparent and understandable results that is in synchronization with the ethical standards and norms of our society. To understand the need of ethical AI systems, we must first understand that a biased or incorrect output from AI system that assist in law enforcement, job recruitment, defense work, self-driving vehicles can result in eroding people's privacy (by misusing data in unintended ways), and taking decisions that are impossible for people to understand and to access liability for damages when harm is caused.

Apart from being **unbiased and accurate** (as shown in **Fig. 8.5**), the credibility of AI systems also depends how we choose to develop and use it. We can just hope that AI will be used for purposes that benefits mankind. But there is nothing that can stop a person from creating systems that can be disastrous for humanity.

In October 2019, researchers found that an algorithm used in US hospitals to predict which patients may require extra medical care heavily favoured white patients over blacks. This bias creeped in because of the fact that white patients paid more bills and availed extra medical facilities for the same medical problem as compared to black.

Amazon's hiring algorithm also suffered from bias related issues. The algorithm was found to be biased against women. This was probably due to the fact that the system saw a greater number of men applicants than women.



FIGURE 8.5 An ethical system must be unbiased and accurate

Nonetheless, efforts are being made by numerous agencies, committees, coalitions and expert groups to ensure that we are on the right track to make AI systems do more to heal than harm. For example, the IEEE Global Initiative on Ethics of Autonomous and Intelligent Systems that put out *Ethically Aligned Design*, a publication from 2016 outlines the guiding standards for developing and administering ethical AI solutions. Therefore, ethical AI aims to *ensure is that AI is doing good for the planet.*

Going further on this, the European Commission High-Level Expert Group on AI identified seven guiding principles for creating Trustworthy AI. These guidelines are as follows:

1. AI systems should support **human autonomy** and decision-making.
2. AI systems should be technically robust and **safe** to use. They must have a fallback plan that can be used when something goes wrong.
3. AI systems should ensure data privacy by protecting user's data, and providing adequate mechanisms to maintain the **quality and integrity** of the data.
4. **AI systems should not work on biased data. The system should be clear, explicable and transparent about its underlying** data and the models.
5. **AI system should be trained with diverse, non-discriminatory and fair data and models** to avoid unfair bias.
6. AI systems should be designed to **benefit** everyone, now and in future. They must focus on s**ocietal and environmental wellbeing** to be sustainable and environmentally friendly.

7. **An AI system should have the accountability** to ensure accurate, unbiased outcomes with utmost responsibility.
8. An AI system must give due importance to data privacy and data security. A system that provides design confidentiality, transparency and security into their AI programs must ensure that data is collected, used, managed and stored safely and responsibly.
9. An AI system must be accountable for its decisions. There must be someone in the organization who will be held responsible for any problematic situation.



Have you ever wondered how much information Google has about you? In an Android phone, Google has your entire Contact List. Google knows the names and numbers of people you talk to. Google knows your location. It knows where you go every day and on holidays. Google knows what email you write to whom. Google has your photographs (Google Photos), knows details about your friends (Google Hangouts). All this is done in lieu of free services it provides. But imagine what big risk it poses to our privacy.

8.1.7 Impact of AI on Jobs

Today, AI is on the tip of everyone's tongue and one of the most in-demand areas of expertise for job seekers. The World Economic Forum has predicted that AI and ML will displace 75 million jobs but would also create 133 million new ones in a few months.

Some of our favourite sci-fi movies have shown that the advent of this technology has created a fear that AI will one day make human beings obsolete in the workforce. As the technology is advancing, many tasks that were once executed by humans have now been automated.

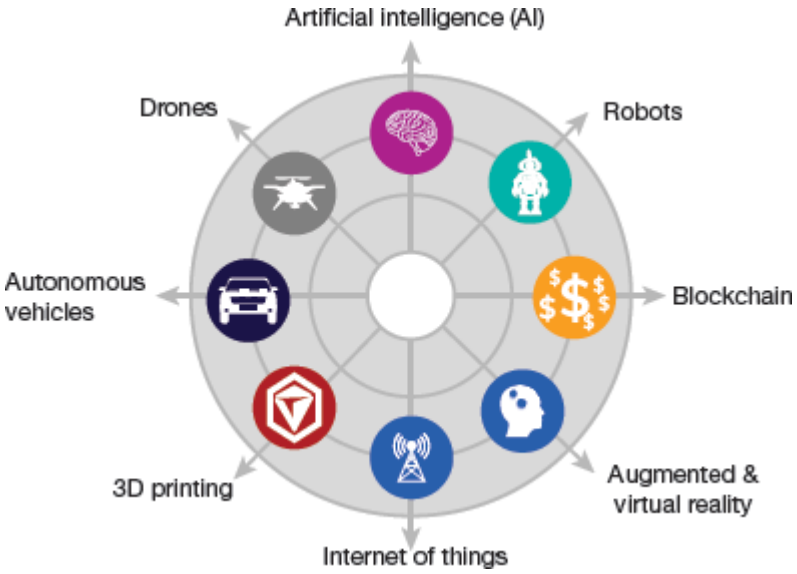


FIGURE 8.6 Emerging Areas in IT

A two-year study from McKinsey Global Institute suggests that by 2030, intelligent agents and robots could replace as much as 30% of the world's current human labour. That is, automation will displace between 400 and 800 million jobs and require approximately 375 million people to switch job categories entirely.

However, job losses due to AI is a fear far from reality. It will just introduce a paradigm shift, similar to the one which occurred after the Industrial Revolution. When the Industrial Revolution took place, many feared that people will lose jobs to machines. But now we see that revolution has generated lot more opportunities and jobs. Similarly, with AI, many professions will become obsolete and disappear, but other new emerging occupations in areas listed in **Fig. 8.6** will become popular. Learning AI skills will help you gain a significant career advantage.

Jobs which will grow with the help of AI include the following.

Creative jobs will get refined and advanced by use of AI. These jobs are usually performed by professionals like artists, doctors and scientists. AI will not only enhance efficiency but also make these jobs less complex for humans.

AI will probably not make human workers obsolete, at least not for a long time.

Management jobs cannot be replaced by artificial managers because anytime, human managers will have to control artificial managers. Managing is a very complex task that involves deep understanding of emotions, profession, ethics, people and communication. Human managers can use certain tools to become more effective at their jobs.

Tech jobs including programmers, data scientists, data analysts, big data engineers and all those who work on the creation and maintenance of AI systems will always be in high demand. However, a few tech jobs which are in demand today may become less common, while others may become more vital.

A few job titles which can be expected to appear by 2030 include Chief Bias Officer, Data detective, Man–Machine teaming manager, AI business development manager, AI assisted medical professional, AI tutor, to name a few.

In many sectors of business, AI has grown by 270% over the last four years. Rather than promoting the obsolescence of humans, AI will continue to drive massive innovation that will fuel many existing industries and could have the potential to create many new sectors for growth, ultimately leading to the creation of more jobs.

We must understand that although AI has done a lot to replicate the efficacy of human intelligence in executing certain tasks, AI programs suffer from major limitations. They can only solve one problem at a time. These systems become rigid, and are unable to think outside their prescribed programming. In contrast, humans have generalized intelligence which helps them solve any kind of problem, think abstractly to make critical judgement across all sectors.

Moreover, AI can learn only with massive amounts of relevant data that is available at the right time. Many a time, this data is protected under issues of privacy and security. In addition, data may have bias. And it is quite obvious that biased data will never give accurate results.

AI Is becoming the standard in all businesses, not just in the world of tech

Machines may also suffer because of availability of limited computation and processing power. For example, the cost of electricity required to power one supercharged language model AI was estimated around $4.6 million.

So, AI has the potential to ultimately create more jobs, not less. The question is not 'humans or computers' but 'humans and computers' involved in complex systems that advance industry and prosperity. Nobody would want to be behind the curve when it comes to AI. In fact, 90% of leading businesses have already invested in AI technologies. More than half of businesses that have implemented some manner of AI-driven technology report experiencing greater productivity.

AI and machine learning top the skills required by companies these days and jobs in this area are expected to increase by 71% in the next five years. AI is likely to have a strong impact on certain sectors in particular. These are discussed below.

1. **Medical:** The medical industry has huge data which can be utilized to create predictive models related to healthcare and by physicians for diagnosis. And at a higher level, AI and automation will also help to eliminate disease and world poverty.
2. **Automotive:** Autonomous vehicles, autonomous navigation and manufacturing within the automotive sector have all been possible with AI technology.
3. **Cybersecurity: During the pandemic, cyber frauds rose by 600%** as hackers capitalized on people working from home, on less secure technological systems and Wi-Fi networks. AI and machine learning provide powerful tools to identify and predict threats. AI is also playing a vital role in the field of finance as it can process large amounts of data to predict and catch instances of fraud.
4. **E-commerce:** On e-commerce websites (like Amazon), AI is used to design chatbots, recommend products, use image-based targeted advertising, and assist in warehouse and inventory automation.

5. **HR applications: An** HR department gets thousands of applications every day. AI tools can be used to reject 75% of resumes that do not fit the current context. Prior to AI, HR managers had to devote considerable time to filter applications to select suitable candidates. Data from LinkedIn shows that recruiters spend up to 23 hours looking over resumes for one successful hire.
6. **Legal profession:** Support services with *document handling -classification, discovery, summarization, comparison, knowledge extraction, and management* use AI for efficiency and accuracy.

Thus, the AI revolution will lead to a new era of *prosperity, creativeness* and *well-being*. Humans will no longer be asked to perform routine, limited-value, full-time jobs. Rather, jobs will be *flexible and offer selective premium services.* There will be more jobs related to programming, robotics, engineering, etc. And, blue-collar and white-collar jobs will be eliminated.

8.2 AI as a Service (AIaaS)

In artificial intelligence as a service (AIAAS), companies use off-the-shelf AI tools to implement and scale AI techniques at a very low price as compared to building a complete in-house AI system.

When a software is provided as a service across the network, it usually uses cloud computing. Similarly, the idea of providing AI services using cloud computing has allowed even small business organizations to use cost effective solutions to improve their performance in a big way.

8.2.1 Factors Triggering Growth of AIaaS

In the last couple of years, AIaaS is growing fast as a business opportunity and many startups have joined to lead the AI revolution. Some major advancements in IT in the last few years have fostered the adoption of AIaaS. Some examples are as follows:

1. Availability of cloud platform with a variety of affordable options for enterprise data management.
2. Data storage technologies have improved and have become cheaper and reliable.

3. Streaming devices and IoT technologies generate massive amounts of data which when analysed can render vital information required to gain competitive advantage.
4. Availability of semi or fully automated data management, analytics and BI products changed the way business was analysed.

Moreover, some major factors that have motivated business to use AIaaS are listed below.

1. Expensive infrastructure
2. Lack of trained programmers
3. Very high charges charged by programmers for implementing AI in a particular organization
4. Many companies lacked sufficient data to analyse

8.2.2 The Growth of AIaaS

With cloud services becoming incredibly accessible, AI has seen a big boom. Now, AI can be made available to a greater number of companies and even those companies can collect and store any amount of data.

Previously, companies were hesitant to build their own clouds to develop, test and utilize their own AI systems. But now with AIaaS, every other company can take advantage of data insights without making huge investment in talent and resources. Moreover, AIaaS has been a lucrative option as business can now easily accomplish the following goals.

1. **Focus** on its core area rather than getting involved in becoming an expert in machine learning.
2. **Reduce operational and infrastructural cost** to a big extent. A complex AI system may require many parallel machines and speedy GPUs. Not many companies can afford such huge investment especially in its initial years. But with AIaaS, every small company can also harness the power of machine learning at significantly lower costs.
3. Minimize the **risk of investment.**
4. **Time** required to deploy a solution is short.
5. **Pre-built algorithms** make complex data management accessible to even small business organizations.

6. Availability of **on-demand services** empower business with a competitive edge.
7. **Increase the benefits** obtained by analysing data trends.
8. **Develop cost effective**, flexible and transparent solution. AIaaS provides a software to the organization that enables them to access machine learning capabilities using templates, pre-built models and drag-and-drop tools to assist developers in building a more customized machine learning framework. Companies no longer have to maintain a team with AI skilled professionals, train them, upgrade their knowledge and spend on areas that only partially support decision-making.
9. **Improve its strategic flexibility.** Companies can only pay for what they actually use. Though machine learning requires a lot of computing power to run, small companies may actually be using only in short amounts that too in spurts. So, in such a scenario, AIaaS is a huge saving.

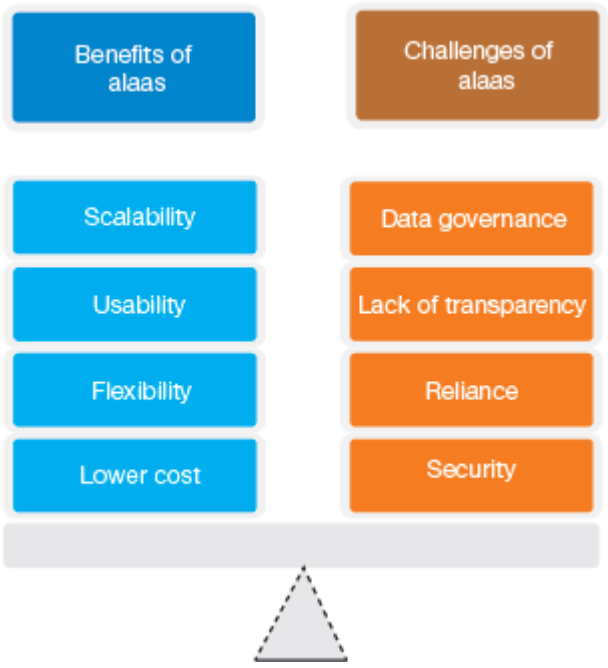


FIGURE 8.7

10. **Scalability is another big advantage of** AIaaS. Now companies can delve into AI technology with smaller projects. If they find it beneficial, then probably, they can experiment more services. At any point in time with the change in demands of the ongoing project, companies can scale up or down their demand for services.

Figure 8.7 summarizes the advantages and shortcomings of providing AI as a Service.

8.2.3 Challenges of AIaaS

1. **Reduced security:** To avail services over cloud, companies have no option but to share their data with third-party vendors. Lot of data security techniques need to be incorporated to ensure that data is safe during storage and transmission and illegal access is strictly avoided.
2. **Reliance;** When working with third parties, the organization becomes dependent on them to satisfy their information needs. This results in delay in getting response.
3. **Reduced transparency:** In AIaaS, organizations are paying to get service, but not the access. Third party vendors take the input and give the output. The entire process works like a black box. The organization has no idea about the inner workings (like algorithms used). This may result in confusion or miscommunication regarding the stability of data or the output.
4. **Data governance:** Every industry and organization has its own data policy. These policies may restrict data storage on a third-party cloud, thereby preventing a company from taking full advantage of AIaaS.

8.2.4 Vendors of AIaaS

There are multiple AI provider platforms offering various machine learning and AI services. Organizations can match their needs with the services and compare the cost at which those services can be accessed to select the offering that is best suited to their current requirements.

Cloud AI service providers allow users to use GPU-based processing for intensive workloads along with their staffing and maintenance staff as well as catering to the hardware changes for different tasks. Some major vendors of AIaaS are Amazon Web Services (AWS), Microsoft Azure and Google Cloud Platform (GCP). Each of these vendors offer different types of bots, APIs and machine learning frameworks to cater to the organization’s demands. Apart from these, countless start-ups are focusing on AIaaS. It is not uncommon for larger companies to purchase smaller companies to add the developed services to their portfolios.

An organization decides on an AI service based on its goals, business size, available budget, technical capabilities of its in-house teams and the

amount of data that needs to be processed.

MonkeyLearn is an AI platform that simplifies text analysis through intuitive, no-code tools. Users can either use pre-trained model like survey analyzer to classify customer feedback by topic or build customized machine learning models to detect sentiment, keywords, and topics in the data. Once developed, these models can be integrated with apps through point-and-click integrations or via the API.

IBM Watson hosts a suite of AI tools and offers many pre-built applications, like Watson Assistant and Watson Natural Language Understand to build virtual assistants and to perform advanced text analysis tasks respectively. Similarly, IBM Watson Studio can be used to build, train and deploy machine learning models across any cloud. With these offerings even users with no expertise on machine learning or data science can make the most of their data.

Microsoft Azure is Microsoft's public cloud computing platform that provides a range of AI and machine learning solutions for developers. For example, Azure Cognitive Services can be used to add computer vision or text extraction capabilities to apps using APIs. Similarly, Azure Bot Service can be used to incorporate any type of bot in the application.

Google Cloud ML is Google's AI platform that basically facilitates data scientists and developers working with big data to create and deploy machine learning projects.

AutoML is being extensively used to train custom machine learning models for text analysis, image classification, translation and more. Data in the datasets can be visualized, 'what-if tools' can be used and metrics can be analysed to assess performance of the model.

AIPaaS or *AI Platform-as-a-Service is an end-to-end solution like a cloud platform, using which business organizations can utilize required services on a pay-per-use or pay-per-service basis. With AIPaaS, users can use third-party APIs that provides a complete, intelligent data management platform.*

Software-as-a-Service (SaaS) market is valued at $133 billion and the advanced technology platforms-as-a-service market globally is forecast to reach about $11 billion by 2023 and surpass '$88,500 million by the end of 2025.'

8.3 Robotics

We have read that robot is a programmable machine that can complete a task. Correspondingly, the term robotics is the study focused on designing, developing and programming physical robots which are able to interact with the physical world and automate one or more tasks. Each robot has a different level of autonomy varying from human-controlled bots that carry out tasks to fully-autonomous bots that perform tasks without any external influences.

8.3.1 Artificially Intelligent Robot

Many people think that AI and robotics are synonymous. But this is not true. **Table 8.1** highlights the difference between an AI program and a robot system.

TABLE 8.1 Difference between AI Program and Robot

AI Programs	Robots
They operate in computer-simulated worlds.	They operate in the real physical world.
Receives input in the form of symbols and rules.	Receives input as analogue signal or in the form of the speech waveform.
To operate an AI program, general-purpose or special-purpose computers are required.	To operate a robot, special hardware with sensors and effectors are needed.

Moreover, from the Venn diagram as shown in **Fig. 8.8**, it is clear that there is one small area where the two fields overlap and, in this area, we have artificially intelligent robots (AIRs). These AIRs are the bridge between robotics and AI.

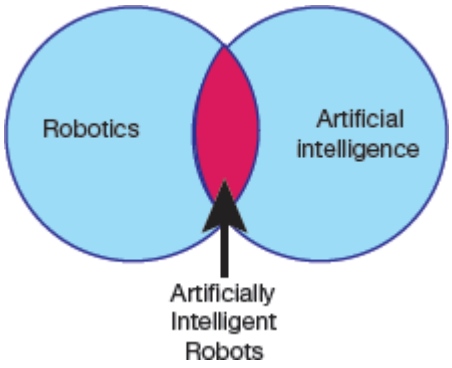


FIGURE 8.8 Venn diagram depiction relationship between AI and Robotics



Credit: PaO_STUDIO / Shutterstock

FIGURE 8.9 AI Robot

Although many AIRs (refer **Fig. 8.9**) are controlled by AI programs, they are not artificially intelligent. For example, most industrial robots are programmed to carry out a repetitive series of movements which do not require artificial intelligence. However, such non-intelligent robots have limited functionality.

To perform complex tasks, an artificially intelligent robot must implement some AI algorithm(s). For example, consider the scenarios given below to understand how AI can guide the functionality of a robot.

Case 1: A warehousing robot may use a path-finding algorithm to navigate around the warehouse.

Case 2: A drone may use autonomous navigation to return to its owner when it is about to run out of batteries.

Case 3: A self-driving car uses AI techniques to detect and avoid potential hazards on the road.

8.3.2 Characteristics of Robots

A robot is a **mechanical device** which completes it tasks in the environment for which it is designed. For example, the Mars 2020 Rover's wheels are motorized and made of titanium tubing to firmly grip the harsh terrain of the red planet.

1. Robots need **electrical components** that control and power the machinery. Usually, they draw electric current using a battery.
2. Robots work on instructions written using **computer programming**. These instructions tell the robot what to do, when to do and how to do it.

8.3.3 Types of Robots

Mechanical bots come in a variety of designs, shapes and sizes to efficiently perform the task they are designed to perform. For example, RoboBee is a 0.2-millimeter-long robot while Vindskip is a 200-metre-long robotic shipping vessel. To classify them broadly into different categories, robots can be divided based on their capabilities to perform a particular task.

Pre-Programmed Robots

They operate in a controlled environment where they perform simple, monotonous tasks. For example, a mechanical arm on an automotive assembly line to weld a door on, to insert a certain part into the engine, etc. (as shown in **Fig. 8.10**) is an example of such a robot that performs its tasks faster and more efficiently than a human.



Credit: Pop Nukoonrat / 123RF

FIGURE 8.10 Pre-Programmed Robots

Humanoid Robots

They look like or mimic human behaviour and usually perform human-like activities (like walking, carrying objects). These days, humanoids look like us. For example, Hanson Robotics' Sophia and Boston Dynamics' Atlas.

Autonomous Robots

These machines operate independently of human operators and are usually designed to perform tasks in open environments without any human supervision. Autonomous robots perceive the world around them using sensors and then use decision-making capabilities to take the optimal next step based on their data and mission. For example, Roomba vacuum cleaner uses sensors to roam freely throughout a home. Other examples of autonomous robots are lawn trimming bots, hospitality bots, autonomous drones and medical assistant bots.

Most industrial robots are non-intelligent.

We often call these robots, cobots. Cobot, or a simple collaborative robot, is a non-intelligent robot. For example, a cobot can be programmed to pick up an object and place it elsewhere. This is done by training a specialized computer vision program to identify different types of objects generally by using an AI algorithm called Template Matching. Broadly, this is an autonomous function as no human intervention is required once the cobot has been programmed to do its task efficiently. Moreover,

the task does not require any intelligence as the cobot will always be picking the object in the same way until its instructions are not changed.

Teleoperated Robots

These are semi-autonomous bots that use a wireless network to enable human control from a safe distance (as shown in **Fig. 8.11**). They are usually deployed in extreme geographical conditions, weather and circumstances. For example, drones used to detect landmines on a battlefield, robots used to fix underwater pipe leaks during oil spill are examples of teleoperated robots.



Credit: MONOPOLY919 / Shutterstock

FIGURE 8.11 Teleoperated Robots

Augmenting Robots

Also known as VR robots, they are used to enhance or replace current human capabilities. With their help, science fiction could become a reality very soon by making humans faster and stronger. For example, robotic prosthetic limbs or exoskeletons are used to lift hefty weights.

8.3.4 Types of Robots Based on Degree of Human Control

Independent Robots

They perform their work autonomously and independent of human operator control. So, they need to be intensely programmed. Such robots are deployed to replace humans for performing dangerous, mundane or otherwise impossible tasks ranging from bomb diffusion and deep-sea travel to factory automation. Though independent robots eliminate certain jobs, the also present new possibilities for growth.

Dependent Robots

These robots are non-autonomous robots as they interact with humans to perform their actions. Humans guide robots to enhance and supplement their already existing actions. For example, these days, advanced prosthetics (the branch of surgery concerned with the making and fitting of artificial body parts. For example, a piece of flexible material applied to a person's face or body to change their appearance temporarily) are controlled by the human mind.

In 2018, Johns Hopkins APL created a popular example of a dependent robot was created by in 2018 for a patient (Johnny Matheny) whose arm was amputated above the elbow. A modular prosthetic limb was fitted in Matheny that was controlled by electromyography, or signals sent from his amputated limb that controls the prosthesis. Over time, he could accurately move his arm and even play the piano. At this time, the signals sent from his amputated limb became smaller and less variable.

Chatbots

Software robotics, also called bots, are computer programs that perform tasks autonomously. For example, a chatbot is a computer program that simulates conversation both online and over the phone and is often used in customer service scenarios. While simple chatbots answer questions with an automated response, more complex digital assistants learn from user information.

Robotics involves building robots physical whereas AI involves programming intelligence.

Types of Bots

Bots, or software robots only exist on the Internet and originate within a computer. Hence, they cannot be called robots. We have learnt in the characteristics of a robot that a robot has a physical form, such as a body or a chassis. Some popularly used bots are as follows:

1. Chatbots carry out simple conversations to provide 24 X 7 customer service.
2. Spam bots collect email addresses and send spam mail.
3. Download bots automatically download software and apps.
4. Search engine crawler bots scan websites and make them visible on search engines.
5. Monitoring bots collect statistics to report speed and status of the website.

8.3.5 Components of a Robot

Robots are created to perform a variety of tasks to present a solution for a wide range of problems. For this, they need a set of specialized components that are discussed below.

Control System

Control system is a CPU that directs a robot's task at high level. It performs all computations to tell a robot how to utilize its specific components, like how our brain sends signals to other parts of the body to complete a specific task. The tasks that a robot completes vary from an invasive surgery to assembly line packing.

Sensors

Sensors are devices that detect the events or changes in the environment and send data to the computer processor. For this, they are equipped with other electronic devices to provide electrical signals that allows the robot to interact with the world. Basically, the electrical signals act as a stimulus that are processed by the controller. Based on the stimulus received, the controller instructs the robot to interact with the outside world. This helps the robot to efficiently respond to real-time data. Popularly used sensors within a robot include video cameras (as eyes), photoresistors (react to light) and microphones (as ears). All these sensors allow the robot to capture information about its surroundings and process it to deduce the most logical conclusion and give commands to other components.

Robots are programmable, autonomous or semi-autonomous machines.

It is through these sensors that the robots get the ability to see, hear, touch and move like humans. *Sensors are devices or machines which help to detect the events or changes in the environment and send data to the computer processor. These devices are usually equipped with other electronic devices*. Similar to human organs, the electrical sensor also plays a crucial role in artificial intelligence and robotics. AI algorithms control robots by sensing the environment, and provide real-time information to computer processors.

Actuators

Actuators are the motor parts that facilitate a robot's movement. We have read that a machine is a robot if it has a movable frame or body and actuators are the components that cause movement. Actuators are made up of motors that receive signals from the control system and move in tandem to carry out the movement necessary to complete the assigned task. They are made of metal or elastic.

Electric motors are the devices that convert electrical energy into mechanical energy and are required for the rotational motion of the machines.

Actuators help in moving and controlling a robot by using energy that can be electrical, hydraulic and air, etc. This means that actuators are usually operated using electricity (electrical), compressed air (pneumatic actuators) or oil (hydraulic actuators). Actuators can create linear as well as rotary motion.

Power Supply

Power supply is done to the robot by a battery. Just as we need food to do our work, robots require power to operate. The main use of power supply is to convert electrical current to power the load. Stationary robots used

in a factory work on AC power while others operate via an internal battery.

While a majority of robots work on lead-acid batteries as they are safe and have long shelf life, others use compact and expensive silver-cadmium batteries. These batteries are chosen depending on safety, weight, replaceability and lifecycle of the robot.

AI algorithms are used in Google searches, Amazon's recommendation engine and GPS route finders.

However, for future robotic development, pneumatic power from compressed gasses, solar power, hydraulic power, flywheel energy storage, organic garbage through anaerobic digestion and nuclear power are also being considered as potential sources of power.

Example of a pure AI is AlphaGo.

Pure AI is usually found in games. For example, in 1997, AI Deep Blue could easily defeat world champion. A more recent example is AlphaGo which won Lee Sedol, the world champion Go player, in 2016.

End Effectors

End effectors are the physical, typically external components of a robot that facilitate it to complete a task. For example, in factories, robots usually have interchangeable tools like paint sprayers, gripping claws or even hands for undertaking tasks like deliveries, packing, bomb diffusion, etc.

All these components are central to every robot's construction.

8.3.6 AI Technology Used in Robotics

In this section, we will list AI technologies that are used in robotics. We have already studied some of them. These technologies are discussed

below.

1. **Computer Vision:** As studied earlier, Computer Vision is an important domain of artificial intelligence that extracts useful information from images, videos and visual inputs to perform appropriate action(s).
2. **Natural Language Processing (NLP):** NLP is used to give voice commands to AI robots to create a strong human-robot interaction. NLP helps the robot to understand and reproduce human language.
   Besides recognizing human language, robots can learn the accent, and predict how humans speak.
3. **Edge Computing:** Edge computing in robots helps in robot integration, testing, design and simulation to provide better data management, lower connectivity cost, better security practices, more reliable and uninterrupted connection.
4. **Complex Event Process:** An event is said to occur when there is a change of state and one or more events combine to define a complex event. Correspondingly, complex event processing (CEP) helps us to understand the processing of multiple events in real time.
   The term ‘complex event process’ is popularly used in various industries including healthcare, finance, security, marketing, detecting credit card fraud, stock marketing, etc. For example, the deployment of an airbag in a car is a complex event that is based on real-time data coming from multiple sensors.
5. **Transfer Learning and AI:** Transfer learning solves a problem with the help of an already solved problem. So, the knowledge gained from solving one problem is utilized to solve another problem that is related to the problem to be solved. For example, the model used to recognize a square shape can also be used to identify a circular shape.
   Since transfer learning reuses the pre-trained model for a related problem, the actual learning time for the new problem to be solved is relatively less and thus provides a cheaper solution. In robotics, transfer learning trains one machine using other machines.
6. **Reinforcement Learning:** Reinforcement learning focuses on learning based on feedback. In this technique, an AI agent to learns and explores the environment, performs actions and automatically learns from experience or feedback obtained for each action. Reinforcement learning makes the agent learn to behave optimally through hit-and-

trail method while interacting with the environment. It is mainly used to deduce decisions to achieve goals in an uncertain and potentially complex environment.

In the context of robotics, robots explore the environment to learn how to work optimally through hit and trial method. For every correct action, the robot gets a reward and a punishment for an incorrect action. In this way, reinforcement learning provides a framework to design and simulate sophisticated and hard-to-engineer robotic behaviours.

7. **Affective computing:** Affective computing focuses on developing systems that can identify, interpret, process and simulate human emotions. Therefore, this study aims to imbibe emotional intelligence in robots so that they are able to observe interpret and express themselves emotionally.
8. **Mixed Reality:** Mixed reality is an emerging field that focuses on programming by demonstration (PbD). PbD creates a prototype using a combination of physical and virtual objects.

8.3.7 Planning and Navigation

In this section, we will read about mobile robots that are critical to robust mobility. For this we must have some knowledge about the kinematics of locomotion, sensors for determining the robot's environmental context and techniques for localizing with respect to its map. Here, we will talk about cognitive level of robots.

Cognition refers to the capability of a system to make purposeful decision and execute them to achieve its highest-order goals. In the context of a mobile robot, cognition refers to robust mobility. Robot navigation aims to make the robot reach its goal position efficiently and reliably based on its knowledge about its environment, the goal position and values of the sensor.

Although many different approaches have been proposed for navigating a robot, there are strong similarities between all of them. The underlying difference exists in the manner in which they decompose the problem into smaller sub-units. Given a map and a goal location, path planning

identifies a trajectory that will help the robot to reach the final destination.

8.3.7.1 Competencies for Planning—Planning and Reacting

Path planning is a strategic problem-solving competency (capability), as the robot must make vital decisions to achieve its goals. The second competency is tactical extreme to avoid all obstacles and collisions by modulating the trajectory of the robot. Given readings of real-time sensor readings, several approaches are proposed to avoid obstacles. We will discover a few of them.

While navigating, a robot plans and reacts. Without reacting, the planning effort(s) will be of no use as the robot never be able to reach its goal. Correspondingly, without planning, the robot will never reach its goal.

Suppose a robot R at time i in the initial belief state b_i has a map M_i to reach position p while satisfying certain some temporal constraints $loc_g(R) = p; (g \leq n)$ (like before timestep n).

Although the robot's movements have to be physical, the robot can only sense its belief state, not its physical location. We will, therefore, map the goal of reaching location p to reaching a belief state b_g. We need a plan q that specifies one or more trajectories from b_i to b_g.

However, the problem arises when either the robot's position is not quite consistent with b_i or M_i is incorrect and/or incomplete. Furthermore, the real-world environment is dynamic. Even if M_i is correct at time i, M may change over time as the robot must incorporate new information gained during plan execution.

Over a period of time, the environment changes and the robot's sensors gather new information. In this scenario, reacting becomes more important. Reacting modulates the robot's behaviour to update the planned upon trajectory so that the robot still reaches its goal position. Moreover, unanticipated new information also requires changes to the robot's strategic plans. Therefore, in the specified time limits, the plan should consider every new piece of information in real time to produce a new

plan to react to the new information appropriately. This theoretical extreme, the point at which the concept of planning and reacting merge, is called integrated planning and execution.

8.3.7.2 Key Terms in Trajectory Planning

Trajectory Planning

Trajectory planning refers to moving from point A to point B while avoiding collisions over time in a 2D or 3D space (refer **Fig. 8.12**). It is an important concept in designing autonomous vehicles. Trajectory planning also known as motion planning is mistakenly referred to as path planning. However, it is different from path planning as it is parametrized by time. Besides path planning, trajectory planning also incorporates planning based on velocity, time, and kinematics.

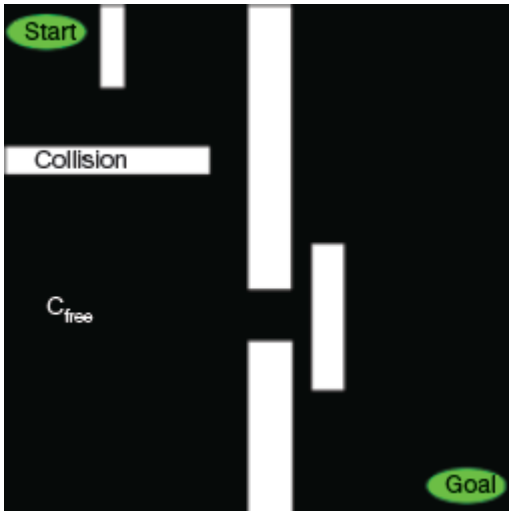


FIGURE 8.12 Trajectory planning

Configuration Space

A configuration is the pose of a robot describing its position. Correspondingly, Configuration Space C, is the set of all configurations. For example, in a 2D space, a robot's configuration is given by coordinates (x, y) and angle θ. Similarly, in 3D space, a robot's configuration is given by coordinates (x, y, z) and angles (α, β, γ).

The goal of path planning is to find a path from the initial position to the goal position in the physical space that avoids all collisions with the ob-

stacles. This problem becomes particularly more difficult as k grows large. If we define the configuration space obstacle O as the subspace of C, the free space in which the robot can move safely can be computed as, F = C - O.

Figure 8.13 shows a picture of the physical space and a 2D configuration space for a planar robot arm with two links. The robot's goal is to move its end effector from position c1 to c2.

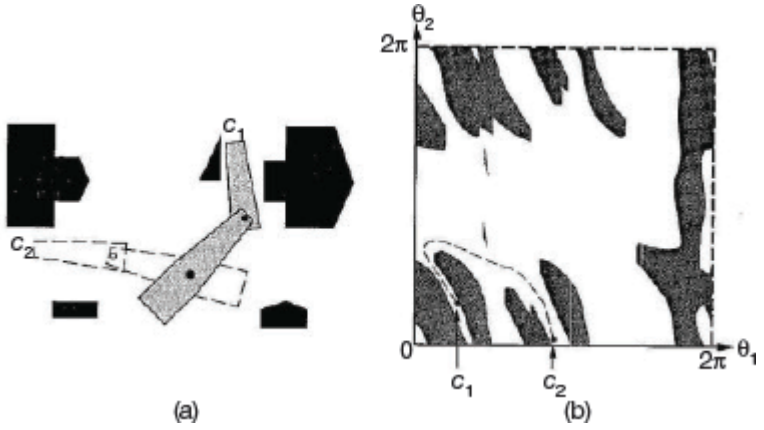


FIGURE 8.13 Physical space and a 2D configuration space

Free Space

Free space C_{free} is the set of all configurations that are collision-free. The robot should be programmed to use kinematics and collision detection from sensors to ensure if a given configuration is a collision free or not.

Target Space

Target space is a linear subspace of free space and consists of space in which we want the robot to explore. In global motion planning, target space is observable by the robot's sensors. But in local motion planning, some states in target space are not observable by robot's sensors. To solve a problem, robots assume several observable virtual target spaces (around itself). The virtual target space is often referred to as sub-goal.

Degrees of Freedom

It specifies modes in which a mechanical device or system can move. This means that the number of degrees of freedom is equal to the total num-

ber of independent displacements or aspects of motion. For example, in a robotic arm, shoulder and wrist can move up or down, left or right but elbow can move up or down. Wrist and shoulder can also be rotated (or rolled).

Therefore, such a robot arm has five to seven degrees of freedom. And if the robot has two arms, then total number of degrees of freedom is doubled.

Multi-legged mobile robots can have more than 20 degrees of freedom. For example, Project Nao, which looks superficially like a large space-age doll, has 25 degrees of freedom.

8.3.7.3 Problem Constraints in Trajectory Planning

Trajectory planning suffers from two main problems which are discussed below.

Holonomicity

This gives the relationship between the controllable degrees of freedom and total degrees of freedom of the robot. A robot is said to be holonomic if the number of controllable degrees of freedom is greater than or equal to the total degrees of freedom. Holonomic robots are easy to work with as they facilitate many movements easier to make. Moreover, returning to a past pose is much easier.

For example, if a self-driving car is designed as a non-holonomic robot, then it will have no way to move laterally, thereby making its certain movements (like parallel parking) difficult. In contrast to this, a holonomic vehicle would be one with mecanum wheels, such as the new Segway RMP.

Dynamic Environments

In dynamic environments, like the real world, objects that may cause collision are not stationary. This makes trajectory planning more difficult as

objects are moving with respect to time. Many-a-time, it may not be possible for a robot to move backward in time. Moreover, many choices are completely irreversible due to terrain, such as moving off of a cliff.

8.3.7.4 Planning Algorithms

In this section, we will discuss three types of planning algorithms.

1. **Artificial Potential Field:** The algorithm places values over the map with the goal having the lowest or highest value that either increases or decreases depending on the distance from the goal (refer **Fig. 8.14**).



FIGURE 8.14 Artificial Potential Field
Obstacles defined can have an incredibly high or low value. The robot is programmed to move to the lowest or (highest) potential value adjacent to it, which should lead it to the goal. However, this algorithm often gets trapped in local minima.

2. **Sampling-based Planning:** Roadmap method is an example of sampling-based planning method. The algorithm first selects a sample of N configurations in C as milestones. Then, a line PQ is formed between all milestones as long as the line PQ is completely in $C_{\mathbf{free}}$. Following this, any graph search algorithms can be used to find a path from start to the goal. Though a large number of N increases computation time, it results in better solutions.
3. **Grid-based Planning:** This algorithm overlays a grid on the map (as shown in **Fig. 8.15**) and then ensures that every configuration corresponds with a grid pixel. The robot can move from one grid pixel to

another adjacent grid pixel if that grid pixel is in C_{free}. T

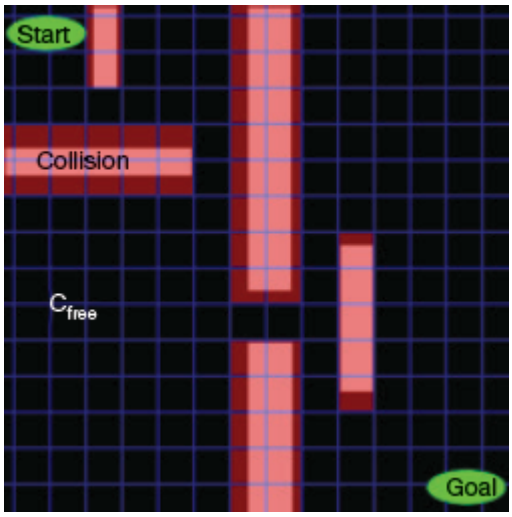


FIGURE 8.15 Grid-based Planning

Search algorithm like A* is used to find a path to reach the goal position from the starting position. Although a lower resolution grid having bigger pixels will make the search procedure execute faster, it may however miss paths through narrow spaces of C_{free}. Moreover, as the resolution of the grid increases, memory usage also grows exponentially. In such a scenario and especially in large areas, another path planning algorithm may be necessary.

4. **Reward-based Planning:** Algorithms based on research-based planning assume that robot in each state (position) may select any action (regarding motion). However, the result of each action is not definite. In such a case, outcomes or displacement, in this context, are partly random and partly under the robot's control. On reaching the goal, the robot gets a positive reward and a negative reward on colliding with an obstacle.

   Research-based planning algorithms, therefore, aim to find a path giving maximum future rewards. Markov decision processes (MDPs) is a popular technique used in many of reward-based algorithms. Though this technique generates optimal path, this path may not be smooth as it limits the robot to choose from a finite set of actions. To overcome this drawback, usually Fuzzy Markov decision processes (FDMPs), an extension of MDPs, is used to generate a smooth path with using a fuzzy inference system.
5. **Road-map Path Planning:** Road-map path planning algorithms capture the connectivity of the robot's free space in a network of one-di-

mensional curves or lines, called road-maps. Once a road-map is constructed, it is then used to plan a robot's motion. The algorithm reduces the problem of path planning to connecting the initial and goal positions of the robot to the road network and then searching the roads from the initial position to the goal position of the robot. This means that, road-map uses the obstacle's geometry to decompose the robot's configuration space.

The challenge is to construct a set of roads that together helps the robot to move around anywhere in its free space, while minimizing the number of total roads. In this section, we will discuss two approaches for road mapping—visibility graph and Voronoi diagram. While visibility graph finds paths that have minimum lengths by bringing the roads closer to the obstacles, Voronoid diagram, on the other hand, keeps roads as far as possible from the obstacles.

8.3.7.5 Visibility Graph

The visibility graph for a polygonal configuration space C includes all edges (including initial and final positions) joining every pair of vertices that can see each other. The unobstructed roads depicted as straight lines in the graph are the shortest distances between two vertices. The job of the path planning algorithm is to find the shortest path from the initial position to the goal position along the roads drawn in the visibility graph (see **Fig. 8.16**).

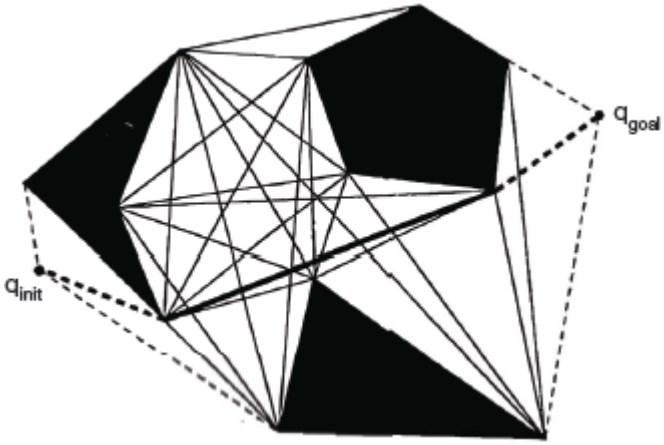


FIGURE 8.16 Visibility Graph

The visibility graph path planning algorithm is very simple and straightforward. It can be readily used when objects in the environment are described as polygons either in continuous or in discrete space. However, there are two important limitations of this algorithm.

First, with increase in the number of obstacle polygons, the number of edges and vertices increases. Therefore, this algorithm works well in sparse environments, but is very slow and inefficient when used in densely populated environments.

Second, the solution paths found by visibility graph planning algorithm takes the robot as close as possible to obstacles on the way to the goal.

Though this algorithm is optimal in terms of the length of the solution path, it does not keep the robot away from obstacles. This compromises safety. A potential solution can be either to grow the obstacle's size significantly so that it becomes more than the robot's radius, or modify the solution path after path planning to distance the path from obstacles. However, such fixes compromise the optimal-length results of the algorithm.

8.3.7.6 Voronoi Diagram

A Voronoi diagram maximizes the distance between the robot and obstacles in the map. For each point in the free space, distance from that point to the nearest obstacle is calculated and plotted as a height coming out of the page (refer to **Fig. 8.17**). As the robot moves away from the obstacle, the height increases. Sharp ridges can be seen in the Voronoi diagram at points that are equidistant from two or more obstacles. Edges are formed by these sharp ridge points. When obstacles in the configuration space are polygons, the Voronoi diagram has straight and parabolic segments as shown in **Fig. 8.18**.

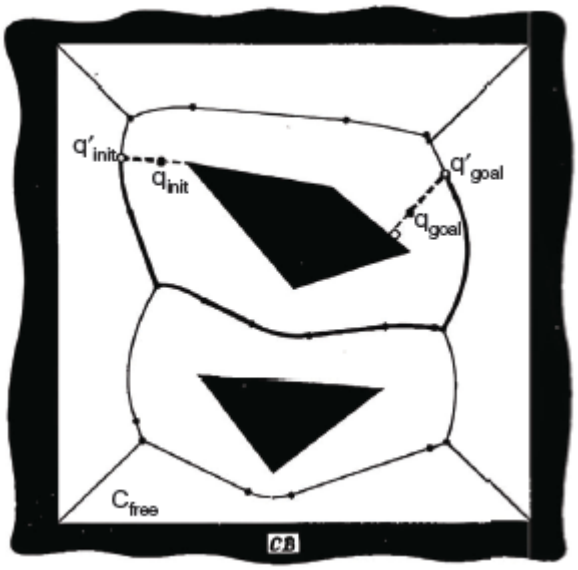


FIGURE 8.17 Voronoi Diagram

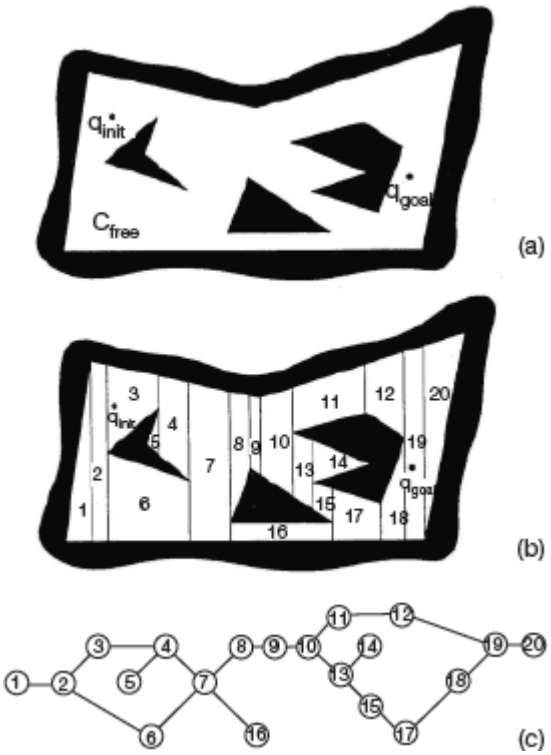


FIGURE 8.18 Forming Edges and making graph from the given Vornoi Diagram

The path in the Voronoi diagram may not be optimal in terms of total path length. Another limitation of this diagram is limited range localization sensors. Since the distance between the robot and obstacle is maximized, any short-range sensor on the robot may not be able to sense its surroundings. Hence, the chosen path may be quite poor from a localization point of view.

However, the main reason of popularity of the Voronoi diagram method is its executability. Given a planned path via Voronoi diagram planning, a robot with range sensors can follow a Voronoi edge in the physical world using simple control rules that match those used to create the Voronoi diagram. Voronoi Diagram can also be used to conduct automatic mapping of an environment by finding and moving on unknown Voronoi edges and creating a consistent Voronoi map of the environment.

8.3.7.7 Cell Decomposition Path Planning

The cell decomposition path planning aims to distinguish between geometric areas, or cells, that are free and areas that are occupied by objects. The following must be kept in mind.

1. The graph is divided into simple, connected regions called ‘cells’.
2. A connectivity graph is constructed by identifying opens cells that are adjacent.
3. Identify the cells in which the initial and goal configurations lie. Select a path in the connectivity graph that joins the initial and the goal cell.
4. From the cells identified, compute a path within each cell following motions and movements along straight lines.

The most important concept in cell decomposition method is the placement of the boundaries between cells. There are two methods which perform the task.

1. *Exact cell decomposition:* In this technique, the boundaries are placed as a function of the structure of the environment in such a way that the decomposition is lossless.
2. *Approximate cell decomposition:* This approach performs decomposition as an approximation of the actual map.

8.3.7.8 Exact Cell Decomposition

In **Fig. 8.19**, exact cell decomposition has been used. Here, the boundary of cells is based on geometric criticality. So, the resulting cells are either completely free or completely occupied resulting in a complete path. In this method, the robot’s ability to traverse from each free cell to adjacent

free cells is more important than the position of the robot within each cell of free space.

Advantage

In environments that are extremely sparse, the number of cells will be small.

Disadvantages

1. The number of cells and computational efficiency of the overall path planning algorithm depends upon the density and complexity of objects in the environment.
2. Due to complexities in implementation, the exact cell decomposition technique is less frequently used in mobile robot applications.

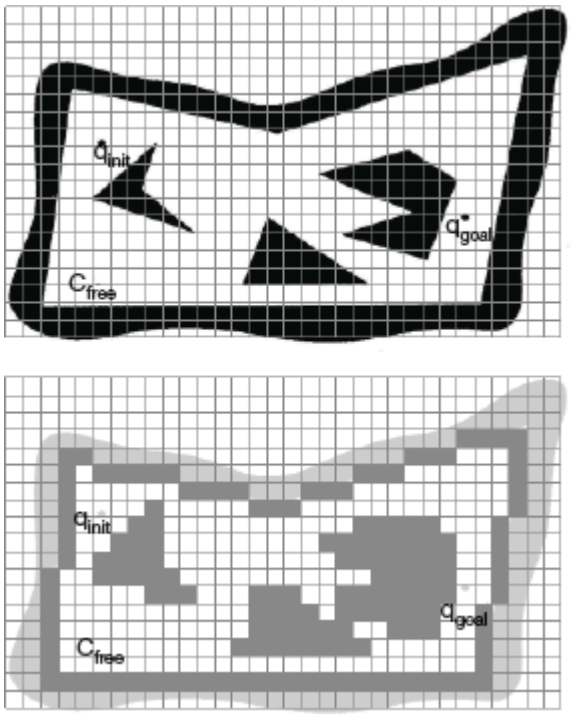


FIGURE 8.19 Exact Cell Decomposition

8.3.7.9 Approximate Cell Decomposition

Approximate cell decomposition is a popular technique for mobile robot path planning. It uses grid-based environmental representations that are themselves fixed gridsize decompositions. They are quite similar to an approximate cell decomposition of the environment.

We can see that in **Fig. 8.20**, the cell size is not dependent on the particular objects in an environment resulting in loss of narrow passageways due to the inexact nature of the tessellation. However, this is rarely a problem as the cell size used is very small, usually 5 cm on each side.

The main advantage of fixed-size cell decomposition is the low computational complexity of path planning. In **Fig. 8.20**, we see that the free space is externally bounded by a rectangle which is further decomposed into four identical rectangles. The rectangle is not decomposed further only in two conditions.

1. First, the interior of a rectangle lies completely in free space.
2. Second, the interior of a rectangle lies completely in the configuration space obstacle.

If the two conditions are not satisfied, then the rectangle is recursively decomposed into four rectangles until one of them is met. Note that the white cells lie outside the obstacles, the black inside and the grey are part of both the regions.

This technique is an efficient and simple strategy for finding routes in fixed-size cell arrays. The algorithm starts from the goal position marking for each cell its distance to the goal cell until the initial robot position is reached. At this point, distance to the goal position and a specific solution trajectory is known by linking together cells that are adjacent and closer to the goal.

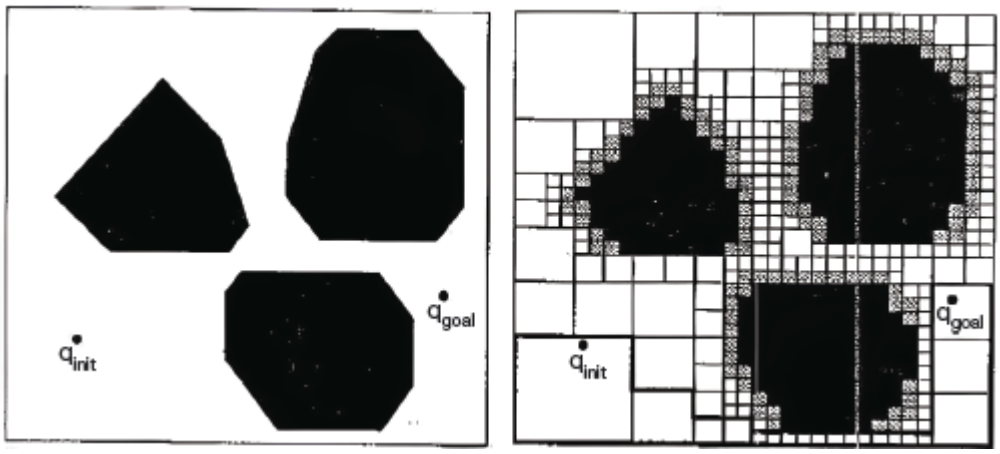


FIGURE 8.20 Approximate Cell Decomposition

If the entire array is in memory, then each cell is visited only once to deduce the shortest path from the initial position to the goal position. So, the search is linear. Therefore, the complexity of this algorithm neither depends on the sparseness and density of the environment, nor on the complexity of the objects' shapes in the environment.

The Cye robot is an example of a commercially available robot that plans path in a 2D space with 2 cm fixed-cell decomposition of the environment. Unlike exact cell decomposition method, approximate cell decomposition method does not guarantee completeness but it is mathematically less involving and easier to implement.

8.3.7.10 Potential Field Path Planning

Potential field path planning algorithm creates a field across the robot's map that directs the robot to the goal position. The potential field technique considers robot as a point under the influence of an artificial potential field $U(q)$.

Like a ball roll downhill under a gravitational field, the robot moves by following the potential field. The goal acts as an attractive force on the robot and the obstacles act as peaks, or repulsive forces. The superposition of all forces is applied to the robot. The artificial potential field thus created helps the robot to avoid known obstacles and smoothly move towards the goal. Thus, this concept goes beyond path planning.

Basically, the resulting field is a control law for the robot. A robot can always decide its next action, assuming that it can localize its position based on the map and potential field.

Knowing that the robot is attracted towards the goal and repulsed by the known obstacle, we need to update the potential field as soon as a new obstacle appear during robot motion. Considering robot as a point, the robot's orientation θ is neglected and the resulting potential field is only two dimensional (x,y).

Assuming a differentiable potential field function U(q), related artificial force F(q) acting at the position (x,y) is calculated as follows:

$$F(q) = -\Delta U(q),$$

where $\Delta U(q)$ is the gradient vector of U at position q.

$$\nabla U = \begin{bmatrix} \frac{\partial U}{\partial x} \\ \frac{\partial U}{\partial y} \end{bmatrix}$$

The potential field acting on the robot is then computed as the sum of the attractive field of the goal and the repulsive fields of the obstacles. Thus,

$$U(q) = U_{att}(q) + U_{rep}(q)$$

Correspondingly, the net force after considering attracting and repulsing forces is calculated as,

$$\begin{aligned} F(q) &= F_{att}(q) - F_{rep}(q) \\ &= -\nabla U_{att}(q) - \nabla U_{rep}(q) \end{aligned}$$

8.4 Recent Trends in AI

In this section, we will read about some latest trends in AI. Extensive research is being done in the terms discussed here.

8.4.1 Collaborative Systems

These days, a lot of research is being done to develop collaborative systems in which humans and machines (with AI) can become effective partners by overcoming each other's limitations and augmenting human abilities. There are many examples of machine-human collaboration as follows.

1. The computer game Foldit uses machine-human collaboration fold simulated proteins. The folding task is performed to understand how real-world proteins involved in the causes of human disease are formed. In the game, both the player as well as AI performs operations. The AI is put to use where it excels and humans' intuition and imagination is applied where they exceed machines.
2. Two amateur chess players and an AI system worked collaboratively and won against a field of supercomputers and grandmasters.
3. In 2014, Knowledge Ventures, a Japanese venture capital company, elected an AI system to its board of directors, to realize the benefits of machine-human collaboration at the highest levels of business.
4. In the military, UAV drones are being used in which a machine takes the front lines with human provides virtual support.
5. The Talos suit in the US military augments a front-line human with a power exoskeleton. The exoskeleton's integrated heating cooling, vital monitoring and heads-up display (HUD) collaborate with humans much similar to that seen in the Ironman movies.

While many people believe that humans and machines can be good partners, others, however, fear that relying too much on AI will result in a massive job loss and making common man people dumb and lazy.

However, the good news is that research involving 1,500 companies says that many companies that have used AI to automate processes and displaced employees will see only short-term productivity gains. Better performance is expected only when humans and machines work together to actively enhance each other's complementary strengths- leadership, teamwork, creativity, social skills of humans and speed, scalability and quantitative capabilities of the machines.

AI can boost our analytic and decision-making abilities and heighten creativity.

Humans Assisting Machines

In a human-machine collaborative system, humans need to perform three important roles. First, they must *train* machines to perform a particular task; *explain* the outcome of that task (especially when results are counterintuitive or controversial) and *sustain* the responsible use of machines so that machines do not harm mankind. For example, the European Union's new General Data Protection Regulation (GDPR) empowers consumers with the right to get an explanation for any algorithm-based decision (like reason for acceptance/rejection of a loan). This feature will not able be a big step in delivering responsible AI but will also create about 75,000 new jobs to administer the GDPR requirements.

Last but not the least, companies also need employees who can continuously monitor to ensure that AI systems are functioning properly, safely and responsibly. For example, a self-driving car must not hit objects and any living being. Similarly, tech companies including Apple use AI to collect personal details about users to improve the user experience, but collecting too much data can compromise privacy, anger customers and even go against the law. Therefore, employees make sure that AI use data only for analysis and the privacy of individual users is not violated.

Points to remember:

Humans must also train AI systems to interact with humans in the best possible way. For example, all virtual assistants were given extensive training to develop to react as a confident, caring and helpful personality without being bossy. Such training requires inputs from a team of experts (even including a poet, a novelist and a playwright).

These days, AI assistants are also being trained to display sympathetic feelings. For example, the start-up Koko, an offshoot of the MIT Media Lab, is developing an empathetic assistant. For example, if a user had a bad day, then Koko would feel sorry for it, ask for more information and offer advice to help the person see his issues in a different light.

8.4.2 Machines Assisting Humans

Smart machines are helping humans enhance their abilities in three ways —amplify, interact and embody. AI systems boost human's creativity, analytic and decision-making abilities by providing the right information at the right time. They can also interact with company's employees and customers in more effective ways. Even at homes, digital assistants are assisting humans to perform their instructions.

SEB, a Swedish bank, uses Aida, a virtual assistant to interact with millions of customers and assist them in opening a bank account, making cross-border payments and interpreting their voice tone (satisfied or frustrated) to provide better service later. In 30% of the cases when the system is not able to resolve an issue, it directs the call to a human customer-service representative and then monitors that interaction to learn how to resolve similar problems in the future.

Case study Question:

Collect information about how Autodesk's Dreamcatcher AI enhances the imagination of even exceptional designers.

Embodying

AI systems are embodied in machines (like robots) to augment the capabilities of a human worker. Using their sophisticated sensors, motors and actuators, these machines can recognize people/objects and work safely in factories, warehouses and laboratories. Such robots also known as cobots handle repetitive actions that require heavy lifting or some dangerous tasks. In such a collaborative environment, humans are free to do other complementary tasks that require dexterity.

For example, Hyundai has extended the concept of cobot with exoskeletons.

Exoskeletons are wearable robotic devices, which adapt to an industrial worker's location in real time to enable him to perform his job with superhuman endurance and strength. Similarly, Mercedes-Benz, uses cobot

arms to customize cars as per user's expectations according to the real-time choices consumers make at dealer's showroom.

8.4.3 Algorithmic Game Theory and Computational Social Choice

These days, (especially enhanced due to pandemic) with increased Internet connectivity and speed, more people prefer to play games like chess, checker, poker and solitaire. These games are played using a clear set of defined rules. So, there was a need to embed algorithms in machines that allowed machines to play ethically according to pre-defined rules. This led to the onset of game theory. In a multi-agent situation (when more than one person is involved in solving a logical problem), game theory is used to choose from a set of options knowing that *our choice would affect the choices of the opponent* and their decision would affect our choices.

Von Newman is credited for the invention of game theory.

We can understand the concept of game theory by studying the five categories.

1. **Cooperative vs Non-cooperative Games:** In cooperative games, participants can establish alliances to increase their chances of winning the game. Correspondingly, in a non-cooperative game, participants cannot form any alliance.
2. **Symmetric vs Asymmetric Games:** In a symmetric game, each participant has the same goal(s). However, their strategy to win may be different. In contrast to this, in an asymmetric game, the participants have different (and at times conflicting) goals.
3. **Perfect vs Imperfect Information Games:** In perfect information games like chess, all the players can see what moves the other player(s) are making. But, in an imperfect information game (like card game), other players' moves are hidden.
4. **Simultaneous vs Sequential Games:** In simultaneous games, multiple players can take actions concurrently. In a sequential game like in

a board game, each player is aware of the other players' previous actions.

5. **Zero-sum vs Non-zero Sum Games:** In zero sum games, a player gains something that causes a loss to other players but in a non-zero sum game, multiple players get the advantages when another player gains.

When using AI techniques in game theory, two concepts are very important—Nash equilibrium and inverse game theory.

Nash Equilibrium

The Nash equilibrium is a condition in which all the players playing the game accept that there is no better solution to the game than the current situation. No player would gain anything after changing their current strategy. So, any new decision made by the players will not result in anything better.

Example: A typical example of Nash equilibrium is the Prisoner's Dilemma. If two criminals get arrested and are kept in confinement without being allowed to communicate with each other, then look at **Fig. 8.21** to understand the possibilities.

- If either of the two prisoners confess that the other committed a crime, the first will be set free and the other will be sent to prison for 10 years.
- If none of them confess, then they will be sent to prison only for 1 year.
- If both confess, then both will be sent to prison for 5 years.

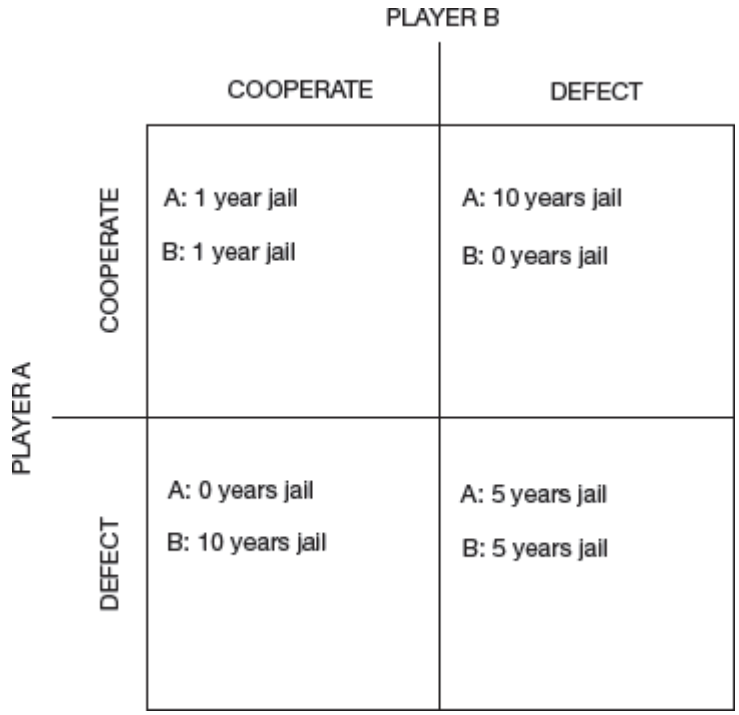


FIGURE 8.21 Nash equilibrium

In this case, the Nash equilibrium is reached when both criminals betray each other.

Inverse Game Theory

Game theory is used to understand the dynamics of a game to optimize the possible outcome of its players. In contrast to this, inverse game theory designs a game based on players' strategies and aims. It is key to designing AI agents environments.

Practical Examples

A generative adversarial network **(GAN)** is a machine learning (ML) model in which two neural networks compete with each other to become more accurate in their predictions.

Nash equilibrium can be attained easily in symmetric than asymmetric games. But, in real world, asymmetric games are more common.

GANs contains two models—a *generative model* and a *discriminative* model (**Fig. 8.22**).

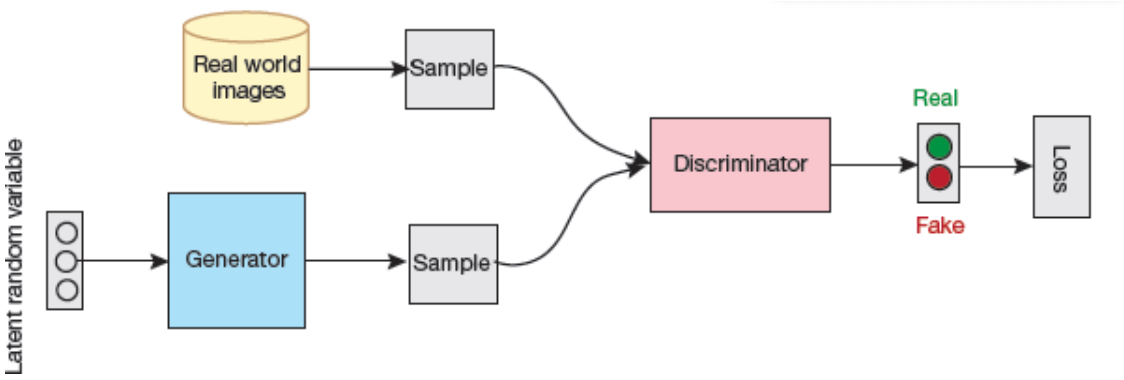


FIGURE 8.22 GAN Architecture [3]

Generative models accept some features as input to examine their distributions and understand how they have been produced.

Discriminative models take the input features to predict the class to which our sample belong.

For example, in a GAN, the generative model uses input features to create new samples that resemble the main features of the original samples.

The original samples and the newly generated samples are then passed to the discriminative model to identify which samples are genuine and which are fake. A common application of GANs is to generate images (as given in **Fig. 8.23**) and then distinguish between real and fake ones.



Credit: doglikehorse / Shutterstock

FIGURE 8.23 Generate Images using GAN Models

Even in game playing, GANs are used extensively. The players (the two models) challenge each other. While one player creates fake samples to confuse the other, the other tries to identify the right samples. The game is played iteratively until Nash equilibrium is reached. In each iteration, the learning parameters are updated to minimize loss.

8.4.4 Multi-Agents Reinforcement Learning (MARL)

Reinforcement learning (RL) is used to make an agent (model) learn through the interaction with its environment. RL was initially developed based on Markov decision processes. This means that the agent placed in a stochastic stationary environment tries to learn a policy through a reward/punishment mechanism. The agent will finally converge to a satisfactory policy.

However, this does not hold true when multiple agents are placed in the same environment. Therefore, instead of depending only on learning through interaction between the agent and the environment, now it is also dependent on the interaction between agents.

Modelling systems with multiple agents is not a trivial task as increase in the number of agents exponentially increases the number of possible ways in which they can interact with each other. Therefore, modelling multi-agents reinforcement learning models can be done using mean field scenarios (MFS) to reduce its complexity by making the assumption a priori that all agents have similar reward functions.

Example: To improve the traffic flow in a city, AI-powered self-driving cars can perfectly interact with the external environment. But things can get complicated if these cars think as a group. In such a case, a car might get in conflict with another when both of them find it most convenient to follow a certain route.

The above situation can be easily modelled using game theory. Our cars would be different players and the Nash equilibrium would be the point between the collaboration between the different cars.

8.4.5 Neuromorphic Computing

Neuromorphic computing is a technique in which a computer (both hardware and software) is modelled based on the architecture of the human brain and its nervous system. It is a science that draws concepts from multiple disciplines including computer science, biology, mathematics, electronic engineering and physics.

The main aim of neuromorphic computing is to create artificial neural systems inspired by biological structures. Such a structure would enable the machine to behave like a human brain in the following ways.

First, to learn, retain information and even make logical decisions as the human brain makes.

Second, to acquire new information and work like a human brain.

Thus, a computer needs to be transformed into a cognition machine.

Did you know that our brain uses roughly 20 watts of power on average, which is about half that of a standard laptop?

How Does Neuromorphic Computing Work?

Traditional neural network and machine learning algorithms are best suited for existing algorithms. They either provide fast computation *or* focus on using low power and there is always a trade-off between the what to achieve. Neuromorphic systems perform fast computation while consuming less power consumption. Additionally, they also resemble human brain as they have the following features:

1. They are massively parallel, to handle multiple tasks at once.
2. They are event-driven, as they respond to events dynamically based on changing environmental conditions.
3. They are economical in terms of power used.
4. They are flexible as they are highly adaptable and are able to generalize.

5. They are strong and fault-tolerant, since they produce results even when some component(s) have failed.

Neuromorphic computing achieves this brain-like function and efficiency by building artificial neural systems that implement ‘neurons’ (nodes that process information) and ‘synapses’ (the connections between those nodes) to transfer electrical signals using analog circuitry. This structure controls the amount of electricity being passed from one node to another to mimic the varying degrees of strength that usually occurs in a human brain. Such a system of neurons and synapses that transmit these electric pulses is known as a spiking neural network (SNN). This facility was not available in traditional neural networks (that uses digital signals).

Neuromorphic systems deviate from the Von Neumann architecture and introduce a new chip architecture that collocates memory and processing together on each individual neuron. Hence, there is no need to have separate memory unit (MU), central processing unit (CPU) and data paths. Having separate parts requires information to be repeatedly moved back and forth between different components to complete a given task, creating a bottleneck for time and energy efficiency. This is known as the von Neumann bottleneck.

By bringing memory on the neuromorphic chip, information can be processed much more efficiently, thereby making the chip more powerful and efficient.

Neuromorphic Computing and Artificial General Intelligence (AGI)

We have seen that the term ‘artificial general intelligence’ (AGI) exhibits intelligence equal to that of humans has not yet been achieved and may not even be reached with the traditional computers. But with neuromorphic computing, researchers are feeling optimistic about it.

Example: The Human Brain Project, featuring the neuromorphic supercomputer SpiNNaker produces a functioning simulation of the human brain and is one of the hottest research projects in AGI.

A machine is said to achieve AGI when the machine can do the following:

- Reason and make judgments in an uncertain situation
- Plan
- Learn
- Communicate using natural language
- Represent knowledge
- Integrate the above skills to fulfil a goal

An AGI may even have the ability to imagine, gain from experience and be self-aware. To confirm whether a system has developed AGI or not, it can be made to pass the Turing Test or the Robot College Student Test, in which a machine enrolls in classes and obtains a degree like a human would.

Debatable Issue

Even if a machine develops human intelligence, then a question rises that whether it should be handled ethically and legally. While some researchers argue that it should be treated as a nonhuman animal in the eyes of the law, others say that still a lot has to be done to inculcate consciousness in machines.

Comparing Human Brain with Neuromorphic Computing

Brains need very less energy than most supercomputers. While a human brain uses about 20 watts, the Fugaku supercomputer needs 28 megawatts (about 0.00007% of Fugaku's) power supply.

Supercomputers need elaborate cooling systems but our brain sits at 37°C in between bones.

Supercomputers can make complex millions and billions of complex calculations in second(s) but the human brain is very adaptable. A single brain can write poetry, identify a familiar face out of a crowd, drive a car, learn a new language, instantly take decisions, respond to events and do much more.

While traditional computers designed on von Neumann architecture are largely serial, brains use massively parallel computing. Brains are also

more fault-tolerant than computers. Therefore, harnessing techniques used by a human brain will be the key to making more powerful neuromorphic computers in the future.

So How Can You Make a Computer That Works Like the Human Brain?

To understand how neuromorphic computers work, let us first recapitulate how the brain works.

When we feel hot or cold, neurons (a type of nerve cell) carry messages to the brain. To transfer messages, several neurons release chemicals across a gap called a synapse. This happens until the message reaches the brain. Once the brain identifies the sensation, it sends an appropriate message to the point which felt the sensation using neurons.

An action potential is triggered either through sending multiple inputs at once (spatial), or by building multiple inputs over time (temporal). The brain transfers information quickly and efficiently through the huge interconnectivity of synapses where one synapse might be connected to 10,000 other synapses.

Neuromorphic computers work like human brains using spiking neural networks (SNN). While transistors used in a conventional computing can be either on (1) or off (0), SNN can produce more than one outputs and transfers information in temporal and spatial way as the human brain does.

Neuromorphic systems can be either digital or analogue. Synapses are controlled either by software or memristors. Memristors can store a range of values instead of storing just one and zero. Like the human brain, they can vary the strength of a connection between two synapses to allow the brain-based systems to learn.

Uses of Neuromorphic Systems

As of now, to perform heavy computational tasks, devices like smartphones hand off processing to a cloud-based system. The cloud processes the query and sends the output to the device. This scenario will change

with neuromorphic systems. In such a system, instead of passing queries and responses back and forth, computations would be done within the device itself.

The current generation AI is rules-based. It is extensively trained on datasets to generate a particular outcome. However, the human brain does not work like this. Researchers are trying hard to make the next generation of artificial intelligence deal with more brain-like problems. For example, constraint satisfaction, in which a system has to find the optimum solution to a problem with a lot of restrictions.

Neuromorphic systems also perform well with noisy and uncertain data.

Examples of Neuromorphic Computer Systems Used Today

64 Intel's neuromorphic chips named Loihi are used to make an 8 million synapse system called Pohoiki Beach, consisting of 8 million neurons (may soon reach 100 million neurons in the near future). They are being used to create artificial skin and also develop powered prosthetic limbs.

IBM's neuromorphic system, TrueNorth, launched in 2014, consists of 64 million neurons and 16 billion synapses. IBM in a partnership with the US Air Force Research Laboratory is trying to create a 'neuromorphic supercomputer' known as Blue Raven for creating smarter, lighter, less energy-demanding drones.

The EU-funded Human Brain Project (HBP), a 10-year project that started in 2013, has led to two major neuromorphic initiatives, SpiNNaker and BrainScaleS. In 2018, the largest neuromorphic supercomputer of that time, a million-core SpiNNaker system went live. It is hoped that this model would soon scale up to one million neurons. BrainScaleS also has a similar goal and its architecture is now on its second generation, BrainScaleS-2.

Challenges to Using Neuromorphic Systems

Shifting from Von Neumann to neuromorphic computing is not that easy and straight-forward. For example, when analysing visual input, tradi-

tional systems treat them as a series of individual frames, but a neuromorphic processor would encode this information as changes in a visual field over time.

New programming languages need to be developed to utilize the change in the underlying architecture. New kind of memory, storage and sensor technology need to be created to take full advantage of neuromorphic devices. This may even change the entire process of creating and integrating hardware and software components.



PROGRAMMER'S ZONE

Lobe is a free, easy-to-use Microsoft desktop application that allow users to build, manage and use custom machine learning models. Once created, these models can be directly uploaded to AI Builder for use with Power Apps and Power Automate.

Step 1: Download and Install Microsoft Lobe.

Step 2: Add and Label Images in Lobe. For this, click the New Project button when Lobe opens. Enter a name for your project. Then add and label some images to create machine learning model.

You can either import images from an existing dataset or capture them from a webcam. While adding images, make sure that you add at least five images of the same subject. Ideally, these should have variations in background, lighting or position to help Lobe identify the important parts.

Now, enter a descriptive label for the first image. Lobe will save this label, so that it can be used for other images in the dataset.

Next, import another set of images of a different but related subject. Label the first image with a descriptive tag, and then use that label to the other images.

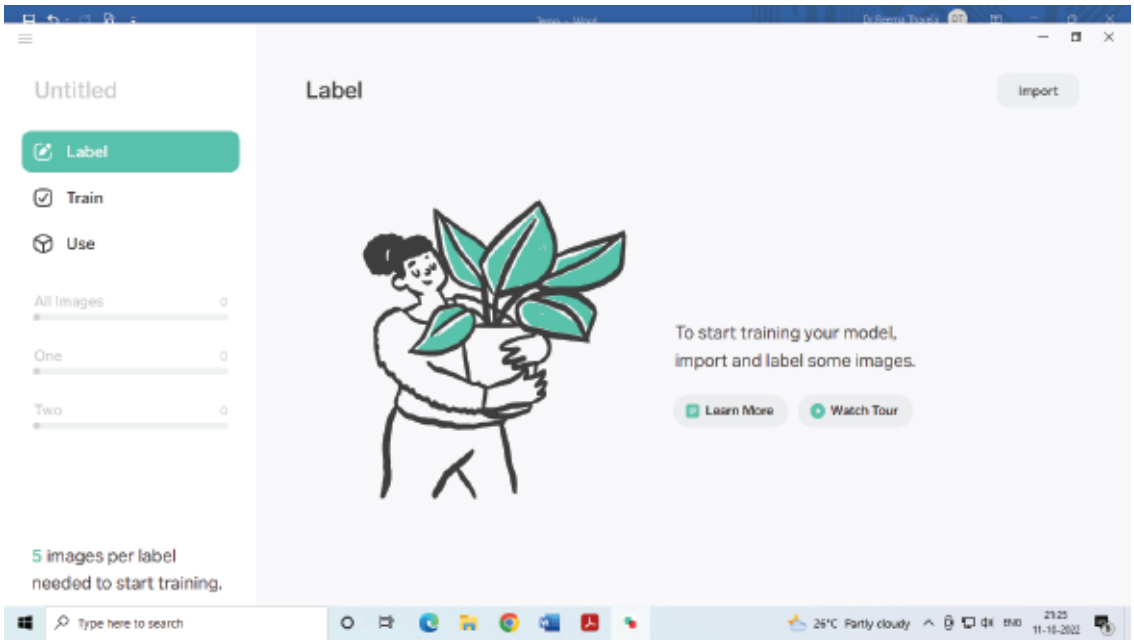
You may add more sets of images. However, to create a model, we need two different sets at least.

Step 3: Train Your machine learning model. You can test if the model works accurately or not by importing another image. See if Lobe predicts the correct label. To check the prediction, click on Play button and then either drag and drop a picture into the app, or click Import. If Lobe correctly predicts the label for the imported image, click the green tick button; if not, click the red button.

You may add any number of images to train the AI model to recognize different variations of the subject. Microsoft suggests that you should add between 100 and 1,000 images per label, depending on the complexity of the task.

Step 4: Optimize the machine learning model. If Lobe makes incorrect predictions, you will have to make it more reliable. You may add more variations of mislabelled images to make future predictions more accurate.

Step 5: Export the Lobe Model to be used in an App.



Credit: Screenshot of Lobe, used with permission from Microsoft



KEY TERMS

Black box AI: The AI program that is not able to explain decisions.

Responsible AI: An AI system that is explainable, monitorable, reproducible, secure, human-cantered, unbiased and justifiable.

Faceswap video blackmailing: Blackmailing people with morphed images and/or videos.

Ethical AI system: An AI system that solves a particular problem, uses unbiased data and is trained using the right learning model for the problem and is monitored to evaluate if the results produced by it are right and fair.

Robotics: The study focused on designing, developing and programming physical robots which are able to interact with the physical world and automate one or more tasks.

Humanoid Robots: Robots that look like or mimic human behaviour and usually perform human-like activities (like walking, carrying objects). Example, Hanson Robotics' Sophia and Boston Dynamics' Atlas.

Autonomous Robots: Robots that operate independently of human operators and are usually designed to perform tasks in open environments without any human supervision.

Teleoperated Robots: Semi-autonomous bots that use a wireless network to enable human control from a safe distance. They are usually deployed in extreme geographical conditions, weather and circumstances.

Augmenting Robots: Also known as VR robots, they are used to enhance or replace current human capabilities.

Independent robots: Robots that are deployed to replace humans for performing dangerous, mundane or otherwise impossible tasks.

Dependent Robots: Robots that are non-autonomous as they interact with humans to perform their actions.

Bots: Also known as software robotics, they are computer programs that perform tasks autonomously.

Chatbot: A computer program that simulates conversation both online and over the phone and is often used in customer service scenarios.

Sensors: Devices that detect the events or changes in the environment and send data to the computer processor.

Actuators: The motor parts that facilitates a robot's movement.

End Effectors: The physical, typically external components of a robot that facilitates it to complete a task. For example, in factories, robots have interchangeable tools like paint sprayers, gripping claws or even hands for undertaking tasks.

Cognition: The capability of a system to make purposeful decision and execute them to achieve its highest-order goals.

Trajectory planning: Moving from point A to point B while avoiding collisions over time in a 2D or 3D space.

Generative adversarial network (GAN): A machine learning (ML) model in which two neural networks compete with each other to become more accurate in their predictions.

Neuromorphic computing: A technique in which a computer (both hardware and software) is modelled based on the architecture of the human brain and its nervous system.



RECAPITULATION POINTS

- An ideal AI system must not be biased.

- As of now, there is no liability for actions on machines. There is no clarity on what legal aspects bind machines when they become increasingly smart.
- In artificial intelligence as a service, companies use off-the-shelf AI tools to implement and scale AI techniques at a very low price as compared to building a complete in-house AI system.
- Stationary robots used in a factory work on AC power while others operate via an internal battery.
- Reinforcement learning focuses on learning based on feedback.
- Affective computing focuses on developing systems that can identify, interpret, process and simulate human emotions.
- Programming by Demonstration (PbD) creates a prototype using a combination of physical and virtual objects.
- Robot navigation aims to make the robot reach its goal position efficiently and reliably based on its knowledge about its environment, the goal position and values of the sensor.
- Path planning is a strategic problem-solving competency (capability), as the robot must make vital decisions to achieve its goals. The second competency is tactical extreme to avoid all obstacles and collisions by modulating the trajectory of the robot.
- Trajectory planning is different from path planning as it is parametrized by time, velocity and kinematics.
- *Degrees of freedom* specify modes in which a mechanical device or system can move.
- A robot is said to be holonomic If the number of controllable degrees of freedom are greater than or equal to the total degrees of freedom.
- Artificial potential field algorithm places values over the map with the goal having the lowest or highest value that either increases or decreases, depending on the distance from the goal.
- Visibility graph finds paths that have minimum lengths by bringing the roads closer to the obstacles; Voronoi diagram, on the other hand, keeps roads as far as possible from the obstacles.
- In collaborative systems, humans and machines (with AI) become effective partners by overcoming each other’s limitations and augmenting human abilities.
- AI systems boost human’s creativity, analytic and decision-making abilities by providing the right information at the right time.

- The Nash equilibrium is a condition in which all the players playing the game accept that there is no better solution to the game than the current situation. No player would gain anything after changing their current strategy.
- Game theory is used to understand the dynamics of a game to optimize the possible outcome of its players. In contrast to this, inverse game theory designs a game based on the players' strategies and aims.
- In a GAN, the generative model uses input features to create new samples that resemble the main features of the original samples. The original samples and the newly generated samples are then passed to the discriminative model to identify which samples are genuine and which are fake.
- Neuromorphic systems deviate from the Von Neumann architecture and introduce a new chip architecture that collocates memory and processing together on each individual neuron.



REVIEW QUESTIONS

1. List some concerns in using an AI system.
2. How can AI become a risk?
3. List the guidelines for making an ethical AI system.
4. Why is AIaaS gaining importance these days?
5. Explain the different types of robots.
6. Explain the different types of bots.
7. Discuss the planning algorithms used by a robot.
8. What do you understand by the term 'degree of freedom'? When is a robot said to be holonomic?
9. Differentiate between two commonly used techniques in road-planning algorithms.
10. Compare the exact cell decomposition with approximate cell decomposition.
11. What are the two types of generative adversarial network?
12. When is a machine said to achieve AGI?
13. Compare the human brain with neuromorphic systems and traditional computers.

14. Give the applications of edge computing.
15. Write a short note on the following:
    1. Collaborative systems.
    2. Algorithmic game theory
    3. Nash equilibrium
    4. Neuromorphic systems
16. Differentiate between:
    1. Human-controlled and fully-autonomous bots
    2. AI programs and robots
    3. Dependent and independent robots
    4. Game theory and inverse game theory
    5. Human brain and neuromorphic systems
    6. Domain adaptation and domain confusion



FILL UP

1. AI program that is not able to explain decisions is known as _____.
2. _____ and _____ ae examples of voice assistants.
3. Some major vendors of AIaaS are _____, _____ and _____.
4. _____ is Google's AI platform used to create and deploy machine learning projects.
5. Software robotics is also called _____.
6. _____ is a computer program that simulates conversation both on-line and over the phone and is often used in customer service scenarios.
7. _____ are the devices that detect the events or changes in the environment and send data to the computer processor.
8. Robots get power supply from a _____.
9. _____ is used to give voice commands to AI robots.
10. _____ planning refers to moving from point A to point B while avoiding collisions over time in a 2D or 3D space.
11. _____ is the pose of a robot describing its position.
12. _____ cell decomposition technique is less frequently used in mobile robot applications.

13. In _____ game of chess, all players can see what moves the other player(s) are making.
14. The _____ is a condition in which all players playing the game accept that there is no better solution to the game than the current situation.
15. _____ is a machine learning model in which two neural networks compete with each other to become more accurate in their predictions.
16. The huge availability of address space in _____ has led to the development of the Internet of Things.
17. _____ learning systems comprise multiple layers and each layer learns different features.
18. In an object detection task, initial lower layers of the network learn _____ features and the higher layers learn _____ features.



STATE TRUE OR FALSE

1. A black-box AI system is explainable, monitorable, reproducible, secure, human-centred, unbiased and justifiable.
2. Super-intelligent AI is unlikely to exhibit human emotions.
3. Cobot is an intelligent robot.
4. Monitoring Bots collect statistics to report speed and status of the website.
5. Actuators can create linear as well as rotary motion.
6. Feedback learning focuses on learning based on feedbacks.
7. Cognition refers to the capability of a system to make purposeful decisions and execute them to achieve its highest-order goals.
8. Holonomic robots are easy to work with.
9. The challenge of road path planning algorithm is to maximize the number of total roads.
10. Path in the Voronoi diagram may not be optimal in terms of total path length.
11. Potential field path planning algorithm creates a field across the robot's map that directs the robot to the goal position.
12. In Non-zero sum games, a player gains something that causes a loss to the other players.

13. Nash equilibrium can be attained easily in asymmetric than symmetric games.
14. Game theory is used to understand the dynamics of a game to optimize the possible outcome of its players.
15. Neuromorphic systems deviate collocates memory and processing together on each individual neuron.
16. The more you want to inherit features from a pre-trained model, the more you have to freeze layers.
17. Transductive transfer learning is used when the domains of the source and target tasks are not exactly the same but are interrelated.
18. *In symmetric feature-representation transfer approaches,* the source features are transformed to match the target ones.
19. Homogeneous transfer learning methods are preferred over heterogeneous ones.
20. In deep neural networks, initial layers freeze and higher layers are fine-tuned.
21. Domain adaptation is done by matching the distributions of the target and source domains to ensure that samples are indistinguishable to the classifier.



MULTIPLE CHOICE QUESTIONS

1. Which type of robot is known as a cobot?
    1. Humanoid
    2. Autonomous
    3. Pre-programmed
    4. Tele-operated
2. _____ are the motor parts that facilitate a robot's movement.
    1. Sensors
    2. Actuators
    3. End effectors
    4. Power supply
3. Actuators are operated using _____.
    1. electricity
    2. compressed air

3. oil
4. All of these

4. Identify the incorrect statement.
    1. An ideal AI system must not be biased.
    2. Teleoperated robots are semi-autonomous bots that use a wireless network to enable human control from a safe distance.
    3. VR robots are used to enhance or replace current human capabilities.
    4. None of these
5. Learning solves a problem with the help of an already solved problem.
    1. Machine
    2. Deep
    3. Transfer
    4. Reinforcement
6. In _____ learning, for every correct action, the robot gets a reward and a punishment for an incorrect action.
    1. Machine
    2. Deep
    3. Transfer
    4. Reinforcement
7. _____ computing imbibes emotional intelligence in robots.
    1. Cognitive
    2. Affective
    3. Edge
    4. Machine
8. Markov decision processes is based on _____ planning algorithm.
    1. Sampling-based
    2. Grid-based
    3. Reward-based
    4. Road-map path
9. In a _____ game, every participant has the same goal(s).
    1. Symmetric
    2. Simultaneous
    3. Zero sum
    4. Cooperative
10. _____ designs a game based on the players' strategies and aims.
    1. Game theory

2. Inverse game theory
3. Nash equilibrium
4. Collaborative systems

11. _____ computing is a technique in which a computer is modelled based on the architecture of the human brain and its nervous system.
    1. Neuromorphic
    2. Cognitive
    3. Affective
    4. Edge
12. Identify the correct statement.
    1. Neuromorphic systems can be either digital or analogue.
    2. Memristors can store a range of values.
    3. Neuromorphic systems also perform well with noise.
    4. Neuromorphic systems work with fixed data and are rule-driven.
13. _____ sensors detect noise and measure distance between two objects.
    1. Speed
    2. Ultrasonic
    3. Radiation
    4. Photo-electric



ANSWERS

Fill Up

1. black box AI.
2. Amazon’s Alexa and Apple’s Siri
3. Amazon Web Services (AWS), Microsoft Azure and Google Cloud Platform (GCP)
4. Google Cloud ML
5. Bots
6. Chatbot
7. Sensors
8. Battery
9. NLP

10. Trajectory
11. Configuration
12. exact
13. Perfect information
14. Nash equilibrium
15. Generative adversarial network (GAN)
16. IPv6
17. Deep
18. Generic, task-specific

State True or False

1. False
2. True
3. False
4. True
5. True
6. False
7. True
8. True
9. False
10. True
11. True
12. False
13. False
14. True
15. True
16. True
17. True
18. False
19. False
20. True
21. False

Multiple Choice Questions

1. b
2. b

3. d
4. d
5. c
6. a
7. b
8. c
9. a
10. b
11. a
12. d
13. b